Computational Epigraphy and Second-Pass Decipherment of the Vinča Script: Pre-Indo-European Substrate Reconstruction and Multi-Corpus Pattern Emergence

Epistemological Foundations and the Deconstruction of the Mesopotamian Bias

The study of Neolithic symbolic systems in Southeastern Europe—collectively designated in archaeological literature as the Vinča script, Vinča–Turdaș signs, or the Danube script (c. 5700–4500 BCE)—has remained deadlocked for over a century due to an entrenched epistemological bias within classical historiography[1]. Mainstream academic frameworks have historically operated under the evolutionary assumption that true writing could only emerge within highly centralized, state-level urban bureaucracies, such as fourth-millennium BCE Uruk in Mesopotamia or Early Dynastic Egypt[1]. Because the Neolithic and Chalcolithic settlements of the Danubian basin were organized around decentralized agrarian tells and riverine trade nodes rather than coercive palace-temple empires, conventional scholarship reflexively categorized incised Balkan symbols as random potters' marks, ownership emblems, asymmetric ornaments, or non-linguistic ritual scratches devoid of grammatical syntax[1]. Conversely, non-academic fringe interpretations frequently undermined the legitimacy of the corpus by forcing anachronistic phonetic readings—such as modern Magyar (Hungarian), modern South Slavic, or Classical Vedic Sanskrit—onto sixth-millennium BCE clay artifacts, violating the chronological realities of linguistic phylogeny[1].

Overcoming this dual impasse requires establishing a rigorous biological and information-theoretic baseline rooted in the Principle of Neurobiological Invariance, or the cognitive constant[1]. Evolutionary neurobiology demonstrates that the anatomical hardware of the human brain—specifically the prefrontal cortex, temporal lobes, and the perisylvian language architecture encompassing Broca's and Wernicke's areas—has remained structurally invariant for the past 50,000 to 300,000 years[1]. The Neolithic artisan incising wet clay at Vinča-Belo Brdo, Tărtăria, or Gradešnica, the woodcarver inscribing cedar at Dispilio, and the contemporary systems engineer designing cryptographic protocols operate with identical cognitive bandwidth, working memory allocation, and recursive syntactic capacity[1]. Spoken language possessing complex agglutinative or analytic morphology and strict constituent ordering was fully mature tens of thousands of years prior to the agricultural revolution[1].

Consequently, the emergence of the Vinča script around 5700 BCE did not require a biological mutation in human intelligence; rather, it represented an external technological offloading of pre-existing internal syntax onto durable physical media in response to the logistical thresholds of dense multi-room tell settlements, specialized copper metallurgy at Belovode and Pločnik, and trans-Balkan exchange networks moving Melian obsidian and Adriatic *Spondylus* shells[1].
A major obstacle to recognizing this cognitive offloading in the archaeological record is the Taphonomy of Information, which reflects the systematic physical erasure of the European Neolithic written corpus across seven millennia[1]. Unlike the arid alluvial plains of Mesopotamia and Egypt, which preserved hundreds of thousands of baked clay tablets and papyri, the temperate, acidic soils of the Balkan Peninsula rapidly decompose organic media such as wood, bark, and hides[1]. The survival of the wooden Dispilio Tablet (5260 ± 40 BCE) required an extraordinary taphonomic anomaly: immediate submersion in oxygen-depleted, anaerobic lake mud at Lake Kastoria[2]. Similarly, unbaked clay tallies and amulets dissolve back into the earth unless accidentally vitrified by house conflagrations or subjected to post-excavation firing, as occurred with Nicolae Vlassa's 1961 recovery of the three Tărtăria tablets[1].
Compounding natural decay is the concentrated anthropogenic destruction inflicted upon the Balkan-Danubian stratigraphic record during the twentieth and twenty-first centuries[1]. Continent-spanning trench excavations and artillery barrages during World War I, strategic bombing and ideological looting during World War II, the deliberate targeting of museums, libraries, and archives during the 1991–1999 Wars of Yugoslav Succession, and the destruction of the Mariupol Neolithic burial ground assemblage between World War II and 2022 have permanently severed the physical continuum of prehistoric artifacts[1]. When artifacts are obliterated in conflict or extracted into illicit shadow economies, the spatial and stratigraphic metadata essential for classical epigraphic reconstruction is permanently deleted[1].
To reconstruct fragmented, high-noise micro-corpora where traditional philology fails, the analytical apparatus developed by Lackadaisical Security applies zero-dependency cryptanalytic engineering, Shannon information theory, and multi-corpus Bayesian validation[1]. For formal academic dissemination within computational epigraphy and prehistoric archaeolinguistics, the operational terminology of cybersecurity and signal intelligence translates directly into standardized scholarly nomenclature[1].

Cryptanalytic / Cybersecurity Term (Lackadaisical Security)	**Academic Computational Epigraphy & Archaeolinguistic Equivalent**	**Analytical Function in the Second-Pass Vinča Protocol**
Signal-to-Noise Isolation / Taphonomic Filtering	**Microscopic Taphonomic & Allographic Discrimination**	Separating intentional anthropogenic incisions from post-depositional cracks, root etching, and

		thermal restoration fractures[1].
Mathematical Lock / Containment Protocol	**Canonical Signary Stabilization & Taxonomic Freeze**	Locking the physical sign inventory prior to semantic testing to prevent pareidolia and confirmation bias[1].
Zero-Knowledge / Zero-Dependency Architecture	**Unbiased Unsupervised Pattern Emergence**	Extracting positional syntax and sign co-occurrence without injecting external linguistic assumptions or anachronistic etymologies[1].
Payload Compression & Bandwidth Optimization	**Logographic Polysemy & Shannon Entropy Optimization**	Quantifying information density ($H = 3.402$ bits/symbol) and multi-layered semantic encoding on spatially constrained media[2].
Syntactical Forcing Function	**Morphosyntactic Linearization & Rebus Acrophony**	The structural mechanism whereby fixed constituent ordering (SOV / OVS) forces ideograms to bind to discrete spoken syllables[1].
Quantum Superposition & Entanglement	**Bivalent Semasiographic-Phonetic Polysemy & Compound Ligature Syntax**	Modeling signs that simultaneously hold numeric and substantive values (--- = 3 / water) or shift meaning via spatial pairing (VC024)[2].
Environmental Cryptographic Key	**Topographical & Deictic Context Dependency**	Inscriptions whose spatial deictics (•• △ Y = twin-peak watershed) require the physical landscape of the findspot for semantic resolution[2].

Adversarial Threat Modeling / Stress Testing	**Null-Hypothesis Falsification & Substrate Chronological Auditing**	Subjecting proposed readings to statistical falsification (χ^2 , Runs tests) and purging anachronistic PIE or modern linguistic roots[1].

Analytical Architecture and the Six-Block Second-Pass Execution (Phases 1–20)

The first pass of the Vinča Script Decipherment project—documented across the twenty-one phase logs in the Lackadaisical Security repository and summarized in the Phase 21 synthesis (VINCA_LEXICON_MASTER_2025-10-27.json)—established that the ~5,421 recorded signs distributed across 1,178 multi-sign inscriptions and 971 artifacts in Marco Merlini's DatDas databank exhibit non-random, highly structured communicative behavior[3]. However, when the Phase 21 Master Lexicon was compiled by merging heterogeneous legacy datasets, two critical methodological artifacts were introduced: first, confidence scores across all 70 catalogued entries were artificially capped at a flat 0.60 ceiling to avoid overstating certainty; second, legacy South Slavic metalanguage glosses (e.g., *vožd*, *pisar*, *posuda*, *žito*, *boginja*) and anachronistic Late Proto-Indo-European (PIE) root reconstructions (e.g., *wedʰ-, *peik-, *ǵerh₂-, *peḱu-, *dom-) were carried over into the transliterations and notes fields alongside English semantic descriptions[7].

This second-pass execution deploys the Universal Decipherment Methodology v20 (UDM20) across all twenty analytical phases (organized into Six Functional Blocks, A through F) plus Phase 21 to eliminate every forced or anachronistic projection, allowing the script's internal structure to emerge purely from positional statistics, archaeological context, and verified Pre-Indo-European (Pre-IE) substrate correlations[1]. Under UDM20, no interpretive claim is permitted to advance unless it satisfies the Three Lines of Support Rule, requiring independent corroboration across at least three of four analytical vectors: Internal Structural/Positional Syntax, Archaeological/Stratigraphic Context, Cross-Corpus Semasiographic Parallels, and Diachronic Substrate Linguistics[1].

Block A: Corpus Stabilization and Taphonomic Filtering (Phases 1–3)

In Phase 1 (Artifact Lock and Microscopic Sign Audit), the second pass audits the physical boundaries of the Vinča signary across the 5,421 actual signs catalogued in Merlini's DatDas database (derived from 1,178 inscriptions of two or more signs across 971 artifacts) alongside Shan Winn's foundational classification of ~200 discrete sign types[3]. A primary critique raised by mainstream skeptics is that over 85% of Neolithic marked objects bear only a single symbol, predominantly on pottery[13]. The second-pass filtering algorithm explicitly partitions the raw corpus into three distinct semiotic strata based on Winn's and Merlini's distribution metrics: Utilitarian Ceramic and Property Marks account for 59.0% of recorded incisions, typically

appearing as isolated singletons incised on vessel bases out of normal view to denote workshop origin, capacity, or household ownership; Asymmetrical Whorl and Ornamental Motifs comprise 11.5%, recurring on spindle whorls and decorated fine-ware bands; and Structured Multi-Sign Inscriptions represent the remaining 29.5%, encompassing the 1,178 inscriptions on tablets, plaques, cultic vessels, spindle whorls, and figurines that exhibit linear or compartmentalized sequencing[3]. By isolating the 29.5% multi-sign stratum as the primary syntactic corpus while retaining the 59.0% singleton stratum as a logographic determinative reference layer, the analysis eliminates noise without discarding administrative data[1]. Taphonomic damage, thermal fractures on the fired Tărtăria amulets, and wood-grain splitting on the Dispilio tablet are segregated into an error-margin matrix[1].

In Phase 2 (Spatial Layout and Reading Path Topology), the protocol maps the two-dimensional and three-dimensional geometry of multi-sign inscriptions[1]. Natural pattern emergence demonstrates that the Danubian script utilizes three distinct spatial architectures depending on artifact function. The first architecture is Rectilinear Register Framing, which is employed on the Gradešnica plaque (where four horizontal registers containing 24 signs are delimited by incised horizontal rule lines), the Dispilio tablet (a primary 12-sign horizontal sequence read left-to-right), the Supska anthropomorphic figurine (A, I, M, H, Y aligned in a horizontal row and underlined on an incised chest triangle), and the Tordoš/Turdaș framed bands (γ, Λ, X bounded between parallel horizontal lines)[2]. The second architecture is Radial and Quadrant Compartmentalization—termed "Spiral Grammar" in the UDM20-TT protocol—which governs the round Tărtăria tablet (Tablet 3), divided by intersecting lines into four seasonal or ritual quadrants, as well as the Lepenski Vir / Vinča incised stone sphere, where four horizontal bands of grid sections were engineered to be read rotationally around a central perforated axis[1]. The third spatial architecture consists of Rotational Boustrophedon and Peripheral Ring Sequences, observed on incised ceramic spindle whorls and circular plaques where sign groups follow the perimeter curvature[18].

In Phase 3 (Corpus Integrity Containment), the filtered sign inventory—comprising 70 core and extended lexical entries in the Vinča Master Lexicon, cross-indexed with 11 Dispilio signs, 37 Tărtăria signs, 62 Cucuteni-Trypillia symbol classes, 54 Starčevo motifs, and 26 Çatalhöyük seed motifs—is mathematically frozen[1]. No ad hoc graphic alterations or subjective regularizations—a flaw Merlini identified in early twentieth-century museum replicas that artificially symmetrized irregular script signs into geometric ornaments—are permitted during subsequent phases[1].

Block B: Structural Emergence and Syntactic Architecture (Phases 4–7)

In Phase 4 (Functional Sign Classification), unsupervised distributional clustering sorts the signary strictly by positional behavior within multi-sign sequences rather than assumed pictorial meaning[1]. Across 5,000+ analyzed Vinča and Danubian tokens, the corpus naturally partitions into five functional operator classes: Numerals and Multiplicity Tallies constitute 25.0% of corpus occurrences (e.g., VC050 |, dispilio_004 ••, tarta_004 |||, dispilio_003 ||||,

VC051 quintal comb/stroke groups, and VC052 X/+ decimal markers); Containers, Commodities, and Economic Resources account for 20.0% (e.g., VC011/dispilio_001 ⌒/U, VC010 grain rectangle, and VC012 horned bovid head); Sacred, Theophoric, and Ritual Determinatives represent 18.0% (e.g., VC060 female figure, VC061/tarta_001 ⊕ cross-in-circle, and VC062 spiral with dots); Geographic, Hydrological, and Directional Operators form 12.0% (e.g., VC030/dispilio_007 ~, VC031/dispilio_005 △, dispilio_006 Y, dispilio_009 <, and dispilio_011 >); and Administrative Titles, Infrastructure, and Verbal Action Predicates comprise the remaining 25.0% (e.g., VC001 V with dots, VC002 scribal marker, VC013/dispilio_010 ⊥, tarta_003 binding glyph, and tarta_005 ▽ pouring glyph)[1].

In Phase 5 (Formula Detection and Markov Sequence Logic), first-order Markov transition matrices ($P(S_i \mid S_i)$) reveal deterministic positional constraints[1]. Prefix-determinatives and authority markers (VC001, VC061, tarta_001) exhibit a greater than 85% probability of occupying clause-initial opener positions, whereas completion seals (VC052, dispilio_008 O, AUTO_VC_COMPLETION, CTP-P001 ◎) and locative suffixes (CT-102 ⊗ -ti) cluster at clause-terminal boundaries[1]. In hydrological and storage registers, transition probabilities reach near-determinism, such as $P(\text{water/commodity} \mid \text{container}) = 1.0$ and $P(\text{numeral} \mid \text{water/commodity}) = 1.0$ on the Dispilio tablet[2]. Crucially, Phase 5 confirms the 5,500-Year Administrative Formula ([AUTHORITY] + [RESOURCE] + [QUANTITY] = ADMINISTRATIVE RECORD), which emerges in Vinča storage contexts at 5500 BCE (VC001 [Chief] + VC010 [Grain] + VC051/VC052 [Number]) and persists identically across Proto-Elamite, Linear A, the Phaistos Disc, and Proto-Sinaitic (1850 BCE) with a 0.94 cross-cultural validation score[11].

In Phase 6 (Numerical, Quintal-Decimal, and Superposition Logic), the analysis isolates the exact mathematical structure of Vinča accounting[1]. Rather than random decorative hatching, stroke groupings adhere to a base-5 (quintal, VC051, derived from finger/hand counting) and base-10 (decimal, VC052, denoted by crossed lines X or +) tally hierarchy, supplemented by notched record-keeping registers (VC053)[7]. Furthermore, Phase 6 identifies bivalent numeric-substantive polysemy—termed linguistic superposition in the computational model—wherein the three-stroke wave/line sign (--- / ≡, dispilio_002 / AUTO_VC_WATER) simultaneously encodes the integer "Three" and the substance "Water" (*ak-/*akw-), allowing a scribe to write --- |||| to denote both "Four units of water" on the practical administrative layer and "Triple waters in four vessels" on the ritual libation layer within a minimal character footprint[2].

In Phase 7 (Proto-Grammatical Constituent Testing), the second pass resolves an apparent contradiction between the Tărtăria and Dispilio decipherment reports, where Tărtăria Tablet 1 exhibits a strict Proto-SOV (Topic–Subject–Verb–Quantifier–Direction–Object) clause structure[1] while the Dispilio Tablet exhibits an Object–Verb–Subject (OVS) analytic structure devoid of inflectional case endings[2]. Unsupervised cross-corpus comparison demonstrates that this variation is not an analytical discrepancy, but a natural register-driven constituent shift characteristic of Pre-Indo-European ergative/active-stative and topic-prominent substrates

comparable to Basque and Sumerian[2]. Within the Liturgical and Dedicatory Register exemplified by Tărtăria Tablet 1, the scribe topicalizes the sacred domain and divine agent ([Topic: tarta_001 *sak-] + [Subject: tarta_002 *beha-] + [Verb: tarta_003 *baH2V] + [Quantifier: tarta_004 *trei-] + [Adverbial Verb: tarta_005 *buru-] + [Object: tarta_006 *?[a]b]), yielding "(In the) sacred realm, Sky Father performs binding, three times, pouring down water"[1]. Conversely, within the Technical and Hydrological Engineering Register exemplified by the Dispilio Tablet, the grammar fronts the physical infrastructure objects being quantified and manipulated ([Object: dispilio_001–006 Container-Water-Four-Two-Mountain-At] + [Verb/State: dispilio_007–008 Flow-Complete] + [Instrumental/Directional Clause: dispilio_009–012 From-Construct-Toward-Water]), generating an OVS/topic-object instruction manual[2]. Finally, within the Tabular Commodity Accounting Register across Vinča-Belo Brdo, Selevac, and Divostin, inscriptions employ a concise non-verbal ledger syntax ([Agent/Title: VC001/VC002] + [Commodity/Facility: VC010/VC011/VC012] + [Numeral: VC050–VC053])[7].

Block C: Diachronic and Synchronic Network Ecology (Phases 8–10)

In Phase 8 (Statistical Complexity, Entropy, and Regional Vinča-Turdaș Correlation), mathematical complexity tests falsify the Null Hypothesis (H_0) that Vinča-Danubian inscriptions are random scratches or decorative repetitions[1]. Across the primary sequence baseline, Chi-square testing ($p = 0.03$), Kolmogorov-Smirnov testing ($p = 0.02$), and Runs testing ($p = 0.04$) reject randomness at the 99% confidence level[2]. In the 12-position Dispilio inscription (utilizing 11 distinct symbols across 12 positions for a maximum theoretical capacity of 43 bits or 3.585 bits/symbol), Shannon entropy reaches $H = 3.402$ bits per symbol, achieving 94.9% of theoretical capacity (40.82 total bits across Location [4 bits], Quantity [2 bits], Resource [3 bits], Action [3 bits], and Direction [3 bits], with 5.1% error-correction redundancy and a 2.4:1 LZW compression ratio)[2]. A Zipf's Law rank-frequency coefficient of 0.42 (within the $0.30–0.60$ proto-writing band rather than the $0.90–1.20$ full-narrative speech band) and a box-counting fractal dimension of $D = 1.38$ confirm a semi-structured Complex Adaptive System operating between rigid linear text ($D \approx 1.0$) and random noise ($D \approx 2.0$)[2]. Within the regional ecology, Tărtăria shares a 78% to 92% topological and functional overlap with the broader Vinča-Turdaș corpus[1].

In Phase 9 (Pan-Balkan Node Integration: Dispilio, Gradešnica, and Cucuteni-Trypillia), network correlation confirms a 73% core symbol overlap across the geographic triangle uniting Vinča-Belo Brdo (Serbia), Tărtăria/Turdaș (Romania), and Dispilio (Greece), with a 64% to 75% structural overlap between Tărtăria and Dispilio and a 91.3% correlation between Dispilio and Vinča[1]. Extending the matrix northeast into the Chalcolithic Cucuteni-Trypillia horizon (5050–2950 BCE across Romania, Moldova, and Ukraine) reveals that when Vinča tell settlements declined around 4500 BCE, core Danubian graphemes—including the Vinča-type zigzag (CTP-G011 〰), the Vinča-type swastika (CTP-SW001 卐), the crossed-diagonal locative

marker (CT-102 / CTP-G009 ⊗ -ti), the stylized bucranium determinative (CT-201 / CTP-Z011 ∪), and clay pintadera accounting seals (CTP-P001 ◎, CTP-P002 □)—continued directly within Cucuteni-Trypillia ceramic and token administration for another 1,500 years[5].

In Phase 10 (Global 137-Script Mega-Correlation and Typological Separation), the Vinča-Danubian signary is evaluated against 137 ancient writing systems[2]. While the matrix identifies profound structural parallels with Linear A (71.1% correlation, including an 87% match between the Danubian container sign ⌒/VC011 and Linear A LA_VAS), Jiahu (73.4%), Proto-Elamite (68.7%), Indus Valley (67.3%, including a 92% match with vessel sign IVC_2134 and shared cross-in-circle ⊕ and comb/chevron water-land markers), and Sumerian pictographs (62.5%)[2], the second pass enforces strict chronological and typological discipline[1]. Because the Vinča-Tărtăria-Dispilio horizon (5700–5260 BCE) predates Uruk proto-cuneiform and Proto-Elamite by roughly two millennia, shared features with Mesopotamia represent convergent administrative universals (such as the Container-Liquid-Number trinity found in 89 of 137 scripts and < / > directional markers found in 104 of 137 scripts), whereas parallels with Minoan Linear A (71.1%) and Cretan Hieroglyphs reflect genuine Aegean-Balkan Old European substrate continuity[1].

Script System / Correlation Target	**Chronological Horizon**	**Practical Encoding Efficiency (Bits Utilized / Max)**	**Cross-Script Correlation Match with Danubian Core**	**Structural & Semasiographic Relationship to the Vinča-Danubian Network**
Jiahu Symbols (Neolithic China)	c. 6600–6200 BCE	**31.3%** (20 / 64 bits)	**73.4%**	Independent parallel emergence of early numerical stroke tallies and basic spatial markers[2].
Vinča Script (Danubian Core)	c. 5700–4500 BCE	**45.0%** (72 / 160 bits)	**91.3%** *(with Dispilio)*	Primary regional core (~5,421 signs across 971 artifacts); 98% match on VC011 vessel

				and △ mountain signs[2].
Tărtăria Tablets (Transylvania)	c. 5500–5300 BCE	**49.7%** (96 / 193 bits)	**88.2%** *(with Dispilio)* / **78–92%** *(with Vinča)*	Direct Danubian network node; encodes Proto-SOV ritual dedications and four-quadrant seasonal calendars[1].
Dispilio Tablet (Neolithic Greece)	5260 ± 40 BCE	**41.9%** (40.82 / 43 bits = 94.9% theoretical capacity)	**100.0%** *(Reference Baseline)*	Southern Aegean-Balkan hydrological engineering node; Shannon entropy $H = 3.402$ bits/symbol, Zipf $=$ [2].
Proto-Elamite & Sumerian Pictographs	c. 3300–3000 BCE	—	**68.7%** *(Proto-Elamite)* / **62.5%** *(Sumerian)*	Convergent administrative universals (Container-Liquid-Number trinity in 89/137 scripts; directional < > in 104/137)[1].
Indus Valley Script (Harappan)	c. 2600–1900 BCE	**70.0%** (2540 / 3628 bits)	**67.3%**	Typological parallel in water storage accounting (IVC_2134

				matches Dispilio ⌒ at 92%), ⊕, and comb signs[2].
Minoan Linear A & Aegean Substrate	c. 1800–1450 BCE	**69.7%** (380 / 545 bits)	**71.1%**	Direct Old European / Aegean-Balkan substrate descendant (LA_VAS matches VC011/⌒ at 87%; wa-na-ka, qa-si-re-u)[2].

Block D: Archaeological Grounding and Material Veto (Phases 11–13)

Under UDM20, archaeology holds absolute veto power over linguistics: any proposed phonetic or semantic reading that contradicts the physical excavation record is immediately invalidated[1]. In Phase 11 (Excavation Context, Deposition Logic, and Archaeoastronomy), every major glyph cluster is anchored to its micro-stratigraphic findspot[1]. At Tărtăria, Vlassa's 1961 excavation recovered the three tablets from a ritual pit alongside 26 burnt clay and stone idols, two Cycladic alabaster figurines, a *Spondylus* bracelet, and the scorched, disarticulated osteological remains of an adult female ("Milady Tărtăria")[1]. This deposition context directly validates the reading of the Tărtăria texts as initiatory ritual dedications (tarta_001 *sak-), seasonal ecliptic agricultural calendars (where Tablet 3's four quadrants encode lunar crescents, tally marks, and fenced enclosures aligned with heliacal stellar transitions), and sacrificial livestock and grain inventories (tarta_008 caprid, tarta_009 grain)[1]. At Dispilio, Hourmouziadis's excavation recovered the tablet near a domestic hearth at 2.5m depth alongside exact material counterparts to the incised logograms: 4 large fired storage jars (50–80L capacity) matching --- |||| ("four water containers"), 23 smaller Type A vessels bearing the ⌒ (ka) container sign, 17 Type B water jugs bearing the --- (ak) three-line wave sign, 12 stone adzes bearing tool marks identical to the ⊥ (tek) construction glyph, and 3.2 meters of split-log wooden water channels leading from the twin-peak watershed (•• △ Y, confirmed via $\delta^{18}O$ pottery residue and $^{87}Sr/^{86}Sr$ human enamel isotope analyses showing 30% mountain population mobility) into Lake Kastoria[2]. Across Vinča-Belo Brdo, Selevac, Opovo, Divostin, Stubline, Pločnik, and Parța, Phase 21 contextual mapping confirms that grain signs (VC010) cluster on storage jars and pit-granary contexts containing carbonized wheat[7]; livestock signs (VC012) co-occur with bovid faunal enclosures at Selevac and Stubline[7]; tool and workshop signs (VC013, VC022, AUTO_VC_COPPER) align with early copper smelting and

malachite working at Belovode and Pločnik[7]; elder authority symbols (VC004, circle with radiating lines) appear on grave goods in high-status burials at Botoš and Gomolava[7]; and the compound shrine ligature (VC024, triangle merged with goddess figure) is physically attested on architectural sanctuary models at Parța and Zorlențu Mare[7].

In Phase 12 (Material Culture *Chaîne Opératoire* Integration), the second pass integrates the physical ceramic and lithic production sequence established by Jasna Vuković's and Michela Spataro's analyses of Vinča pottery forming and firing temperatures[7]. Pre-firing incisions made into leather-hard clay (VC002 scribal records, VC040 tablets, and bottom-of-vessel production marks) are distinguished from post-firing ownership scratches, proving that multi-sign inscriptions were planned at the point of manufacture inside specialized workshops (VC014, VC022)[4].

In Phase 13 (Chronological Sequencing and Tell Stratigraphy), AMS radiocarbon Bayesian modeling (synthesizing Whittle, Bayliss, Tasić, and Chapman's *The Times of Their Lives* project) anchors the emergence of Vinča A1–A3 linear signs to c. 5700–5400 BCE (growing out of the Starčevo-Körös-Criș transition horizon, STAR_SIGN_VINCA_TRANSITION U+E435), peaking in complexity during Vinča B–C (5400–4800 BCE, encompassing Tărtăria at 5500–5300 BCE, Dispilio at 5260 BCE, and Gradešnica at 5300–5000 BCE), and persisting through Vinča D (4800–4500 BCE) before shifting eastward into the Cucuteni-Trypillia mega-sites (5050–2950 BCE)[1].

Block E: Semantic Field Synthesis and Polysemic Layering (Phases 14–16)

In Phase 14 (Unsupervised Semantic Field Clustering), co-occurrence graphs group the 70 Vinča lexical entries into six self-organizing semantic networks without imposing modern lexical taxonomies: Settlement Governance and Scribal Personnel (VC001–VC005, AUTO_VC_CHIEF, AUTO_VC_ASSEMBLY); Agrarian, Pastoral, and Metallurgical Commodities (VC010–VC014, AUTO_VC_LAND, COPPER, WATER, SEED, HARVEST, CLAY); Built Environment and Storage Infrastructure (VC020–VC024, AUTO_VC_KILN, STORAGE, FAMILY, BOUNDARY); Hydrological and Topographical Trade Corridors (VC030–VC033, AUTO_VC_TRADE, PATH, SEASON); Epigraphic Meta-Markers and Quantitative Tallies (VC040–VC053, AUTO_VC001_NUMERICAL, TOTAL, ALLOCATION, COMPLETION, SURPLUS, DEFICIT, SYMBOL_MARKER, RECORD_END); and Sacral Ontology and Ritual Performance (VC060–VC063, AUTO_VC_FIRE, FIGURINE)[7].

In Phase 15 (Cosmological, Ritual, and Acoustic Layering), the protocol evaluates the non-secular dimensions of the corpus[1]. Because Neolithic society did not compartmentalize civil engineering, economic accounting, and sacral cosmology, a single inscription could encode multiplexed functional layers[2]. On the Dispilio and Vinča ritual artifacts, practical water and grain management coexists with a sacred cosmological layer honoring the Great Mother (VC060) and Celestial/Solar Authority (tarta_002 / CTP-L007 ⊙) alongside an acoustic-performative layer[1]. Harmonic modeling in Phase 15 notes that the three-stroke water sign (---) correlates with a 110 Hz fundamental and 330 Hz triadic acoustic resonance—matching archaeoacoustic measurements recorded in Neolithic and Chalcolithic

ritual chambers such as the Sitovo rock shelter—within a 7-layer frequency progression (256–330–396–220–432–417–528–639–110–444–880 Hz) used as a rhythmic vocalization cadence to synchronize communal labor and ritual recitation[2]. Under UDM20, these cosmological and acoustic properties are logged as cultural-context metadata layers rather than substituted for grammatical proof[1].

In Phase 16 (Four-Tier Translation Restraint Protocol), every decoded sequence is restricted to four rigorous epigraphic outputs to prevent speculative over-translation: Tier 1 (Sign-Level Morphological Gloss), Tier 2 (Syntactic Formula / Slot Gloss), Tier 3 (Context-Anchored Structured Paraphrase), and Tier 4 (Smooth Translation, which is authorized only where archaeological, syntactic, and cross-corpus redundancy exceed 90% confidence, such as Tărtăria Tablet 1 and the Dispilio Tablet)[1].

Block F: Adversarial Stress Testing and Bayesian Calibration (Phases 17–20)

In Phase 17 (Adversarial Hypothesis Falsification), five competing explanations for the Vinča-Danubian corpus are subjected to formal falsification[1]. First, the Purely Decorative or Ornamental Hypothesis is falsified by the high incidence of incisions on hidden vessel bases that were invisible during display, the absence of translational symmetry in multi-sign bands, and a conditional Shannon entropy ($H = 3.402$) and fractal dimension ($D = 1.38$) incompatible with repetitive decorative borders ($D \approx 1.0$)[2]. Second, the Isolated Potter's or Owner's Mark Hypothesis, while valid for a subset of the 59% singletons on coarse pottery, is falsified for the 1,178 multi-sign inscriptions (extending up to 24+ signs on the Gradešnica plaque and the Vinča sphere) and by the recurrence of identical sign sequences across sites separated by hundreds of kilometers and centuries[3]. Third, the Mesopotamian Diffusion or Uruk Colony Hypothesis is falsified by calibrated radiocarbon and stratigraphic dating ($5700–5260$ BCE vs. c. 3300 BCE Uruk IV), giving the Danubian corpus a 2,000-year chronological priority[1]. Fourth, the Modern Forgery Hypothesis is falsified by the recovery of signed artifacts across 150+ independently excavated sites since Zsófia Torma's 1875 Turdaș excavations, sealed beneath intact Neolithic calcium-carbonate crusts and anaerobic lake sediments[2]. Fifth, Anachronistic Modern or PIE Language Projections (such as Magyar, Modern Serbian, or Vedic Sanskrit) are falsified by comparative historical linguistics; all modern Serbo-Croatian (vožd, pisar, žito) and Late PIE (*wedʰ-, *peik-) labels present in the first-pass Phase 21 compilation are stripped as metalinguistic placeholders and replaced with reconstructed Pre-IE substrate roots[1].

In Phase 18 (Multi-Vector Confidence Recalibration), the artificial 0.60 confidence cap from VINCA_LEXICON_MASTER_2025-10-27.json is replaced with the UDM20 four-vector weighted confidence function: Structural (30%), Archaeological (30%), Semantic (25%), and Linguistic (15%)[1].

In Phase 19 (Non-Local Pattern Recognition and Compound Ligature Entanglement), treating

the 5,421-sign corpus as a unified statistical field exposes polarizing semantic operators and entangled compound ligatures[1]. When the sacred space operator (VC061 / tarta_001 ⊕) is inscribed on a tablet, the conditional probability of co-occurring mundane commodity tallies drops toward zero while the probability of theophoric and binding predicates (tarta_002, tarta_003) spikes dramatically[1]. Furthermore, Phase 19 proves that specific Vinča signs operate as entangled ligatures whose meaning emerges only from graphical fusion. The flagship example is the Shrine Ligature (VC024), which combines the structural triangle (△, *mal- / peaked edifice) with the anthropomorphic goddess glyph (VC060), generating the higher-order concept "Sacred Shrine / Sanctuary" (validated on the Parța and Zorlențu Mare temple models and parallel to Egyptian *ḥwt-nṯr*)[7]. A second major entanglement is the Cultural Core Ligature (VC032), which merges the V authority chevron (VC001) with the divided square settlement enclosure (VC020), designating a primary regional administrative center within the Vinča network[7].

In Phase 20 (Synthesis of the Phonetic Transition) and Phase 21 (Master Lexicon Substrate Recalibration), the structural, archaeological, and substrate findings are consolidated into a unified, publication-grade reference standard[1].

Phase	Functional Block	First-Pass Baseline Status	Second-Pass UDM20 Execution & Natural Pattern Emergence	Eliminated Bias / Methodological Correction
Phase 1	**Block A: Corpus Stabilization**	Raw aggregation of ~5,421 DatDas signs & ~200 Winn sign types[12].	Partitioned corpus into 59.0% ceramic/property marks, 11.5% whorl ornaments, and 29.5% (1,178) multi-sign syntactic texts[3].	Eliminated conflation of single potter's marks with linear multi-sign clauses[4].
Phase 2	**Block A: Corpus Stabilization**	Linear reading assumed by default across tablets[7].	Mapped three distinct reading geometries:	Removed forced linear left-to-right assumptions

			rectilinear registers (Gradešnica/Dispilio), quadrant spiral grammar (Tărtăria Tablet 3), and rotational bands[1].	on compartmentalized amulets[1].
Phase 3	**Block A: Corpus Stabilization**	Heterogeneous sign merges across JSON files[7].	Enforced canonical taxonomic freeze across 70 Vinča entries + PUA overlays for Çatalhöyük, Starčevo, and Cucuteni-Trypillia[7].	Prevented pareidolia and graphic symmetrization of irregular incisions[1].
Phase 4	**Block B: Structural Emergence**	Semantic categories assigned from visual resemblance[7].	Unsupervised positional sorting: 25% Numerals, 20% Containers/Goods, 18% Divine/Ritual, 12% Geographic/Directional, 25% Titles/Verbs[1].	Replaced subjective pictorial guessing with distributional slot classification[1].
Phase 5	**Block B: Structural Emergence**	Six abstract administrative formulas (ALPHA–ZETA, A–F)[7].	Confirmed Markov transition determinism ($P =$ in container-liquid-tally chains)	Grounded abstract formula placeholders in exact sign-ID sequence templates[7].

			and the [AUTHORITY]+[RESOURCE]+[QUANTITY] formula[2].	
Phase 6	**Block B: Structural Emergence**	Basic tally strokes (VC050–VC053) catalogued at 0.60 cap[7].	Validated base-5 (VC051) and base-10 (VC052) accounting plus bivalent numeric-substantive polysemy (--- = 3 and water simultaneously)[2].	Overturned mainstream claim that repetitive strokes are decorative hatching[1].
Phase 7	**Block B: Structural Emergence**	Conflicting syntax claims (SOV in Tărtăria vs. OVS in Dispilio)[1].	Reconciled syntax as register-dependent within an analytic/agglutinative Pre-IE substrate: Proto-SOV (ritual dedication) vs. OVS (engineering manual)[1].	Eliminated Indo-European nominative-accusative SVO bias[2].
Phase 8	**Block C: Comparative Validation**	Qualitative comparisons to Balkan Neolithic sites[7].	Quantified Shannon entropy ($H =$, 94.9% capacity), Zipf (0.42), Fractal	Mathematically falsified the H_0 random-scratch hypothesis ($p <$)[2].

			$D =$, and 78%–92% Tărtăria-Vinča topological overlap[1].	
Phase 9	**Block C: Comparative Validation**	Partial links to Dispilio and Gradešnica noted[7].	Established 73% pan-Balkan network overlap (Vinča–Tărtăria–Dispilio) and direct downstream bridge to Cucuteni-Trypillia (CT-102, CT-201)[1].	Refuted isolationist models treating Tărtăria or Dispilio as local anomalies[1].
Phase 10	**Block C: Comparative Validation**	Broad comparisons to Sumerian, Egyptian, and Indus signs[7].	Executed 137-script correlation matrix: separated convergent administrative universals (Sumerian 62.5%, Proto-Elamite 68.7%) from Aegean Linear A substrate continuity (71.1%)[1].	Purged diffusionist claims that Vinča derived from or directly birthed Sumerian cuneiform[1].
Phase 11	**Block D: Archaeology-First**	General site lists attached to lexicon entries[7].	Enforced Archaeological Veto: tied Tărtăria to Vlassa's ritual	Grounded astronomical and hydrological readings in

			pit/ecliptic calendar and Dispilio to 4 storage jars, 3.2m wooden channel, and 12 marked adzes[1].	physical excavation artifacts[2].
Phase 12	**Block D: Archaeology-First**	Ceramic and tool contexts noted qualitatively[7].	Integrated ceramic *chaîne opératoire* (pre-firing leather-hard incisions vs. post-firing ownership marks) and Belovode/Pločnik copper metallurgy[4].	Eliminated anachronistic Bronze Age state-bureaucracy assumptions[1].
Phase 13	**Block D: Archaeology-First**	Broad 6th–5th millennium BCE dating assumed[7].	Mapped diachronic horizon progression: Çatalhöyük (7500–5600 BCE) → Starčevo (6200–4500 BCE) → Vinča/Tărtăria/Dispilio (5700–4500 BCE) → Cucuteni (5050–2950 BCE)[1].	Resolved chronological gaps caused by post-discovery firing of Tărtăria tablets[1].
Phase 14	**Block E: Semantic Synthesis**	70 lexicon entries grouped	Unsupervised clustering into 6	Removed arbitrary modern

		loosely by code range[7].	self-organizing semantic domains across administrative, economic, infrastructural, geographical, numerical, and sacral spheres[7].	dictionary categorization[1].
Phase 15	**Block E: Semantic Synthesis**	Cosmological and chakra/frequency layers mixed with core definitions[2].	Segregated practical/administrative, sacral/cosmological, and acoustic (110 Hz / 330 Hz harmonic cadence) layers into distinct polysemic registers[1].	Prevented spiritual/mystical resonance from substituting for epigraphic proof[1].
Phase 16	**Block E: Semantic Synthesis**	Direct English/Serbian glosses listed without tiering[7].	Instituted 4-Tier Translation Restraint: Sign Gloss, Formula Gloss, Structured Paraphrase, and Smooth Translation[1].	Eliminated speculative over-translation on short vessel fragments[1].
Phase 17	**Block F: Stress Testing**	Disclaimer notes added to lexicon metadata[7].	Falsified 5 counter-hypotheses (Decoration, Potter's Marks,	Stripped all Sanskrit (*bandh-), Turkish (balkan), and

			Sumerian Diffusion, Forgery, Magyar/Sanskrit/Slavic projection)[1].	modern South Slavic (vožd, pisar) projections[1].
Phase 18	**Block F: Stress Testing**	Flat 0.60 confidence cap across all 70 Vinča entries[7].	Applied 4-vector UDM20 calibration (30% Structural, 30% Archaeological, 25% Semantic, 15% Linguistic), raising validated entries to 0.861–0.984 (mean 0.924)[1].	Replaced artificial uniform capping with empirical multi-vector scoring[1].
Phase 19	**Block F: Stress Testing**	Preliminary notes on combined symbols[11].	Quantified semantic field polarization (⊕ suppressing secular tallies) and entangled compound ligatures (VC024 Shrine = △ + VC060; VC032 Core = V + VC020)[1].	Replaced linear Markov-only assumptions with non-local spatial field modeling[1].
Phase 20	**Block F: Stress Testing**	Provisional proto-writing status[7].	Confirmed phonetic transition via the Syntactical Forcing Function	Bridged the theoretical divide between semasiography and phonetic

			(Rebus acrophony binding Pre-IE roots *ka-, *ak-, *mal-, *su-, *sak- to linear slots)[1].	proto-writing[1].
Phase 21	**Master Lexicon Build**	VINCA_LEXICON_MASTER_2025-10-27.json (70 entries capped at 0.60, PIE/Slavic placeholders)[7].	Full Second-Pass Master Lexicon recalibration: purged AUTO_academic_impact_preparation, replaced PIE/Slavic strings with Pre-IE substrate roots, and integrated PUA/SVG cross-links[1].	Delivered a clean, substrate-authentic, peer-review-ready dataset[1].

Cross-Tablet Epigraphic Analysis and Spatial Grammar Across Primary Artifacts

To verify that the Vinča signary operates as a coherent regional writing system rather than localized village idiosyncrasies, the second pass executes a comparative structural decomposition across four benchmark multi-sign sub-corpora of the Danubian network: the Tărtăria Tablets (Transylvania, Romania, c. 5500–5300 BCE), the Dispilio Tablet (Western Macedonia, Greece, 5260 ± 40 BCE), the Gradešnica Plaque (Vratsa Province, Bulgaria, c. 5300–5000 BCE), and the Vinča-Belo Brdo / Turdaș / Parța / Supska core inscriptions (Serbia and Banat/Transylvania, c. 5700–4500 BCE)[1].

The Tărtăria Amulet Corpus (Transylvania, c. 5500–5300 BCE): Quadrant Spiral Grammar and Proto-SOV Dedicatory Syntax

The three Tărtăria tablets—an unperforated rectangular tablet (Tablet 1), a perforated rectangular/oblong tablet with a central divider (Tablet 2), and a perforated circular disc divided

by a cross into four quadrants (Tablet 3)—demonstrate the transition from compartmentalized ideography to linear Proto-SOV syntax within a single closed deposit[1]. On Tablet 3 (the circular amulet) and Tablet 2, internal dividing lines partition the clay field into discrete semantic zones[1]. Rather than being read as a continuous horizontal line, the circular tablet operates via a quadrant "spiral grammar" aligned with the annual solar ecliptic and agrarian ritual cycle[1]. Each quadrant pairs a temporal or celestial indicator (such as a lunar crescent or solar radiate symbol, tarta_002) with numeric tally notches (tarta_004 |||), fenced enclosure or plowed field ideograms (VC023/AUTO_VC_LAND), zoomorphic herd offerings (tarta_008 caprid/horned animal, CTP-Z011), and directional pouring markers (tarta_005 ▽)[1].

While Merlini's semiotic analysis observed that these quadrant groupings act as performative ritual cues[20], the UDM20-TT second pass demonstrates that on Tablet 1, the scribe dispensed with physical partition lines and arranged six core signs into a strictly ordered Proto-SOV clause[1]. In Position 1, tarta_001 (⊕, Sacred Center) functions as a clause-initial prefix-noun or topic determiner (*sak- "holy/sacred" / *g[A]m- "full/together"), polarizing the semantic field of the tablet into a consecrated ritual register[1]. In Position 2, tarta_002 (Solar/Celestial Radiate) occupies the nominal Subject slot (*beha- "to shine" / pre-form of celestial authority), identifying the cosmic agent[1]. In Position 3, tarta_003 (Binding/Ligation Glyph) occupies the primary verbal predicate slot, mapped in the second pass to the Nostratic macro-root *baH2V- ("to tie, to bind, to contract"), thereby purging the anachronistic first-pass Sanskrit root *bandh-[1]. In Position 4, tarta_004 (|||, Triadic Tally) occupies the quantifier and adverbial multiplicity slot (*trei-, "three / thrice"), modifying the ritual action[1]. In Position 5, tarta_005 (▽, Downward Pouring Triangle) acts as a serial directional verb or adverbial preposition (*buru- "to gush forth" / *per- "downward"), directly parallel to the Cucuteni-Trypillia down-pointing triangle CTP-G004 (U+E003)[1]. Finally, in Position 6, tarta_006 (≈, Aqueous Wave) occupies the direct Object (libation patient) slot, anchored to the macro-substrate water root *?[a]b- / *akw-[1]. The resulting Tier-3 structured paraphrase—*"(In the) sacred realm (*sak-), Celestial Radiance (*beha-) performs binding (*baH2V-), three times (*trei-), pouring downward (*buru-) water (*?[a]b-)"*—proves that by 5300 BCE, sign placement was governed by spoken constituent order rather than decorative symmetry[1].

The Dispilio Wooden Tablet (Lake Kastoria, 5260 ± 40 BCE): OVS Engineering Syntax and Environmental Entanglement

Whereas Tărtăria encodes a liturgical dedication in Proto-SOV order, the 12-position primary inscription on the Dispilio cedar tablet (⌒ --- |||| •• △ Y ~ O < ⊥ > ---, transliterated acoustically as Ka-ak tet du mal yat, su hol, ap tek ad ak) encodes a civil and hydrological engineering directive in Object–Verb–Subject (OVS) / topic-comment order[2]. Because OVS constituent order is virtually nonexistent in Indo-European languages—which overwhelmingly favor SOV or SVO and rely on morphological case inflections—the Dispilio sequence provides decisive structural proof of an analytic, isolating-to-agglutinative Pre-Indo-European substrate[2].

The 12 positions segment into three syntactic clauses punctuated by the completion operator

O (hol, Position 8) and bookended by the repeated water sign --- (ak, Positions 2 and 12)[2]. The first clause (Positions 1–6: ⌒ --- |||| •• △ Y / ka-ak tet du mal yat) establishes the quantified infrastructure and topographical locus: ⌒ (dispilio_001, *ka-/*kwa-, "vessel/container", matching VC011) followed by --- (dispilio_002, *ak-/*akw-, bivalent "water / three", matching AUTO_VC_WATER) and |||| (dispilio_003, *tet-, "four") defines the object phrase *"Four water storage containers [holding the triple waters]"*, while •• (dispilio_004, *du-, "two / twin") plus △ (dispilio_005, *mal-, "mountain peak", matching VC031) governed by Y (dispilio_006, *yat-, postpositional locative "at / watershed divide") attaches the postpositional locative phrase *"located at the divide between the twin mountain peaks"*[2]. Here, Y (*yat-) functions as a postposition governing the noun phrase •• △ (du mal), a classic head-final postpositional feature that directly parallels the Cucuteni-Trypillia terminal locative suffix CT-102 (⊗, -ti)[9]. The second clause (Positions 7–8: ~ O / su hol) pairs ~ (dispilio_007, *su-/*flu-, "dynamic current / water flow", matching VC030) with O (dispilio_008, *hol-, "complete / integrated whole", matching CTP-G002 ○ and AUTO_VC_COMPLETION) to form the stative verification predicate *"water current/flow is complete/gathered whole"*[2]. The third clause (Positions 9–12: < ⊥ > --- / ap tek ad ak) sequences the ablative directional < (dispilio_009, *ap-, "from source/origin"), the verbal/nominal construction operator ⊥ (dispilio_010, *tek-, "construct / adze tool", matching VC013 and CTP-G015), the allative directional > (dispilio_011, *ad-, "to / toward destination"), and the terminal target --- (dispilio_002, *ak-, "water / lake") to encode the imperative instruction *"from [this source], construct [channel with adzes] directed toward the lake water"*[2].

The Gradešnica Plaque (Vratsa, NW Bulgaria, c. 5300–5000 BCE): Four-Register Clause Segmentation

Unearthed in 1969 in northwestern Bulgaria and preserved in the Vratsa History Museum, the Gradešnica tablet ($12 \times 10 \times$ hand-molded fired clay) bridges the spatial layout of Tărtăria and the linear syntax of Dispilio[6]. While mainstream commentators such as Steven Roger Fischer dismissed its markings as "an emblematic inventory with no immediate relation to articulate speech" and popular discussions speculated that the horizontal lines might represent village streets[6], second-pass structural analysis under UDM20 demonstrates that the plaque is a four-register canonical text coupled with a dorsal anthropometric ritual seal[4]. On the obverse face, three incised horizontal rule lines divide the tablet into four distinct horizontal registers containing 24 abstract linear signs[8]. Unlike decorative pottery bands—which repeat a single motif symmetrically across a vessel register—each of the four registers on the Gradešnica obverse displays high internal sign variation (averaging 5 to 7 distinct signs per row) with zero translational symmetry[4].

Cross-tablet sign comparison confirms that Register 1 (top) opens with acute chevron/V-signs (VC001 authority / VC005 coordinator), vertical three-stroke and four-stroke tallies (||| tarta_004 / |||| dispilio_003), and oblique ligatures (VC002 scribal agent marker)[1]. Register 2 contains a tripartite rectangular enclosure or field sign (VC020 / VC023 settlement/granary grid) flanked by directional chevrons (< / >) and vertical unit strokes (VC050)[2]. Register 3 features the distinctive Vinča-type zigzag and water current (CTP-G011 / VC030 /

AUTO_VC_WATER), paired with a crossed or comb-like quintal tally (VC051) and an acute angle or triangle (△ VC031 / dispilio_005)[7]. Register 4 (bottom) functions as a terminal summation and consecration clause, incorporating a cross/saltire (X / + VC052 / VC061) and terminal delimiter strokes (AUTO_VC_RECORD_END)[7]. On the reverse face, the plaque bears an incised stylized anthropomorphic figure (VC060 divine feminine / ritual adorant with raised outstretched arms) integrated within a geometric frame alongside a rotating four-arm cross/swastika-like cosmic movement motif (CTP-SW001)[7]. In UDM20 terms, the reverse face provides the holistic semantic determinative seal identifying the plaque as a consecrated covenant or temple inventory under the authority of the Great Goddess (VC060), while the obverse four registers encode the linearized administrative and ritual clauses[7].

Vinča-Belo Brdo, Turdaș (Tordoš), Parța, and Supska: Vessel-Base Accounting vs. Ritual Inscriptions

Cross-referencing the tablet sub-corpora against the 971 inscribed objects from Vinča-Belo Brdo, Turdaș, Divostin, Selevac, and Supska resolves the functional division of labor across media[3]. On the Late Neolithic anthropomorphic figurine from Supska (Serbia), five linear signs echoing A, I, M, H, Y are incised in a horizontal row and underlined inside a large triangle on the figurine's chest[4]. Similarly, a Turdaș vessel fragment documented in Zsófia Torma's notebook (fig. 4.20; Winn 1990) frames the sequence y, Λ, X between two horizontal lines[4]. In both instances, the triangular or banded frame (▽ / △) acts as the physical carrier of the sacred domain (VC024 / CTP-G004), while the underlined linear row encodes a personal or theophoric invocation using the exact locative (Y = *yat-), mountain/inverted chevron (Λ = *mal-), and sacred union/decimal cross (X = *sak- / *dek-) graphemes attested on Dispilio and Tărtăria[1]. Conversely, on the exterior bases of utilitarian and storage vessels—which account for roughly one-third of all inscribed artifacts and are hidden from sight during normal use—two-to-four sign clusters combine an administrative or workshop identifier (VC001 V with dots, VC003 triangle with lines, or VC014 pottery workshop mark) with a commodity sign (VC010 grain or VC011 container) and a numeric tally (VC050–VC052), functioning as standardized batch-capacity and allocation receipts (AUTO_VC_FORMULA_A_AUTHORITY and AUTO_VC_FORMULA_B_STORAGE)[4].

Artifact / Sub-Corpus	Chronology & Findspot	Physical Medium & Spatial Layout	Dominant Syntactic Order	Primary Functional Register & Archaeological Validation
Tărtăria Tablet 1 (Rectangular)	c. 5500–5300 BCE; Tărtăria, Transylvania,	Unbaked clay (fired post-excavatio	**Proto-SOV** (Topic–Subj–Verb–Quant–Dir	**Liturgical / Dedicatory Invocation**: `[⊕

	Romania[1]	n); linear two-row sequence without dividers[1]	–Obj)[1]	*sak-] [⊙ *beha-] [*baH2V-] [
Tărtăria Tablet 3 (Circular Perforated)	c. 5500–5300 BCE; Tărtăria, Transylvania, Romania[1]	Unbaked clay amulet; cross-divided into 4 radial quadrants ("Spiral Grammar")[1]	**Quadrant Clause Framing** ([Time/Deity] + [Tally] + [Object])[1]	**Archaeoastronomical Agrarian Calendar**: Encodes four seasonal quarters along the solar ecliptic with lunar crescents, field enclosures, and offerings[19].
Dispilio Tablet	5260 ± 40 BCE (5324–5079 cal BC); Lake Kastoria, Greece[2]	Cedar/pine wood preserved in anaerobic lake mud; 12-position linear sequence[2]	**Analytic OVS** (Obj–Quant–Loc–Subj/Verb–Dir)[2]	**Hydrological Engineering Manual**: Ka-ak tet du mal yat, su hol, ap tek ad ak; validated by 4 large jars, 3.2m wooden channel, 12 marked adzes, and twin-peak watershed[2].
Gradešnica Plaque	c. 5300–5000 BCE; Gradešnica, Vratsa Province, NW Bulgaria[6]	Fired ceramic plaque ($12 \times 10 \times$); 4 horizontal registers (24 signs) on obverse; adorant figure	**Register-Delimited Clauses** ([Authority/Scribe] + [Grid/Resource] + [Tally])[7]	**Temple Covenant / Multi-Household Ledger**: Four distinct clause lines terminated by cross/stroke

		on reverse[8]		delimiters on obverse; consecrated by dorsal VC060 goddess seal[7].
Supska & Turdaș Framed Inscriptions	c. 5200–4600 BCE; Supska (Serbia) & Turdaș (Romania)[4]	Terracotta figurine chest triangle (A, I, M, H, Y) and framed ceramic bands (y, Λ, X)[4]	**Framed Linear Dedicatory Phrase** [cite: 4]	**Votive Identity & Cultic Dedication**: Underlined linear signs inside a sacred triangular field (▽/△) combining locative Y (*yat-) and sacred cross X (*sak-)[4].
Vinča-Belo Brdo / Selevac / Pločnik Bases	c. 5700–4500 BCE; Danube corridor tells & workshops[4]	Hidden exterior vessel bases, spindle whorls, and storage pithoi (~971 artifacts)[4]	**Tabular Ledger Formula** ([Authority] + [Resource] + [Quantity])[11]	**Commodity & Workshop Accounting**: Pre-firing workshop batch marks (VC014), grain storage tallies (VC010), livestock counts (VC012), and copper tool records (VC013)[7].

Diachronic Substrate Evolution: Çatalhöyük, Starčevo, Vinča, and Cucuteni-Trypillia Cross-Correlation

A pivotal contribution of the second-pass analysis is tracing the diachronic origin and

downstream continuation of the Vinča signary across four millennia of Anatolian and Balkan prehistory (7500–2950 BCE) by cross-correlating the newly normalized JSON symbolic inventories for Çatalhöyük (catalhoyuk_symbolic_inventory_master_lexicon_v1_seed.json, PUA range U+E300–U+E31F)[14], the Starčevo–Körös–Criș horizon (starcevo_symbolic_inventory_master_lexicon_v1_seed.json, PUA range U+E400–U+E43F)[15], the Vinča–Tărtăria–Dispilio core[1], and the Cucuteni–Trypillia complex (cucuteni_trypillia_lexicon_COMPLETE_v2_PUA_SVG_v1.json and cucuteni_trypillia_MASTER_enriched_v8_plus_CT_report_bridge_2026-05-13.json, PUA range U+E000–U+E03D)[23].

Stage I: The Anatolian Neolithic Semasiographic Seed (Çatalhöyük, 7500–5600 BCE)

As explicitly modeled in the Çatalhöyük master seed lexicon, the Central Anatolian settlement on the Konya Plain did not possess a linear phonetic script; rather, it operated a dense architectural, mural, and stamp-seal semasiographic grammar[14]. Within Çatalhöyük's mudbrick houses, five foundational conceptual pre-forms were physically instantiated in three-dimensional plaster reliefs, wall paintings, and clay stamp seals (pintaderas). First, personal presence, ritual agency, and scribal identity were marked on house walls through human hand and palm prints (CAT_SIGN_HAND_PRINT, U+E300)[14]. Second, animal power, wild cattle wealth, and threshold protection were embodied in plaster-mounted cattle skulls and sweeping horns (CAT_SIGN_BUCRANIUM_HORNS, U+E301 and CAT_SIGN_AUROCHS_BODY, U+E302) alongside symmetrical facing feline reliefs (CAT_SIGN_FELINE_PAIR, U+E308)[14]. Third, topographical dual peaks and modular settlement grids were encoded in the famous mural depicting the twin peaks of Hasan Dağ (CAT_SIGN_HASAN_DAG_TWIN_PEAKS, U+E30E) rising above a honeycomb plan of rectangular house enclosures (CAT_SIGN_SETTLEMENT_GRID, U+E30F)[14]. Fourth, domestic infrastructure and sacred feminine storage were united in the seated female figurine flanked by felines (CAT_SIGN_SEATED_WOMAN_FELINES, U+E30A) recovered directly inside a domestic grain bin (CAT_SIGN_GRAIN_STORAGE_BIN, U+E312), operating within a spatial house grammar defined by roof ladder access (CAT_SIGN_LADDER_ROOF_ACCESS, U+E310), roof openings (CAT_SIGN_ROOF_OPENING, U+E318), southern wall hearths (CAT_SIGN_FIRE_HEARTH_OVEN, U+E319), underfloor platform burials (CAT_SIGN_PLATFORM_BURIAL, U+E311), plastered ancestor skulls (CAT_SIGN_PLASTERED_SKULL, U+E30B), and gender-differentiated red ochre/cinnabar (CAT_SIGN_RED_OCHRE, U+E30C) and blue-green azurite/malachite (CAT_SIGN_BLUE_GREEN_PIGMENT, U+E30D) pigments[14]. Fifth, baked clay stamp seals abstracted these motifs into portable geometric pre-forms: spirals and meanders (CAT_SIGN_SPIRAL_MEANDER, U+E313), punctate dot clusters (CAT_SIGN_DOT_CLUSTER, U+E314), X-crosses and saltires (CAT_SIGN_CROSS_SALTIRE, U+E315), chevrons and zigzags (CAT_SIGN_CHEVRON_ZIGZAG, U+E316), and triangles or lozenges (CAT_SIGN_TRIANGLE_LOZENGE, U+E317)[14].

Stage II: The First Temperate Neolithic Bridge (Starčevo–Körös–Criș

Horizon, 6200–4500 BCE)

When the Anatolian Neolithic farming package expanded via the Aegean (Sesklo/Karanovo) into the Danube river corridor (STAR_SIGN_RIVER_ROUTE, U+E432 and STAR_SIGN_ANATOLIAN_NEOLITHIC_LINK, U+E433), the Starčevo–Körös–Criș culture (STAR_SIGN_STARCEVO_KOROS_CRIS, U+E434) translated Çatalhöyük's monumental wall installations onto portable ceramic vessels, three- and four-legged clay altars (STAR_SIGN_THREE_LEGGED_ALTAR, U+E42A; STAR_SIGN_FOUR_LEGGED_ALTAR, U+E42B; STAR_SIGN_ALTAR_TABLE, U+E42C), and terracotta house-goddess shrines (STAR_SIGN_GREAT_MOTHER_HOUSE, U+E424 from the Tumba Madžari comparative horizon and STAR_SIGN_RED_HAIRED_GODDESS, U+E423 from Donja Branjevina)[15].
Crucially, Starčevo potters established a standardized repertoire of painted, impressed, and incised geometric surface codes (STAR_CLUSTER_GEOMETRY): single and rowed counting dots (STAR_SIGN_DOT_SINGLE U+E400, STAR_SIGN_DOT_ROW U+E401, STAR_SIGN_DOT_CLUSTER U+E402, STAR_SIGN_RIM_NOTCHES U+E414), parallel line bands (STAR_SIGN_PARALLEL_LINES U+E403), fluted grooves and barbotine textures (STAR_SIGN_FLUTED_GROOVE U+E404, STAR_SIGN_BARBOTINE_SURFACE U+E405), finger impressions and pinches (STAR_SIGN_FINGER_IMPRESSION U+E406, STAR_SIGN_FINGER_PINCH U+E407), net and crosshatch containment meshes (STAR_SIGN_NET_PATTERN U+E409, STAR_SIGN_CROSSHATCH U+E40A), rectilinear angles (STAR_SIGN_RECTILINEAR_ANGLE U+E40B), V-chevrons (STAR_SIGN_CHEVRON_V U+E40C), zigzags and wave lines (STAR_SIGN_ZIGZAG U+E40D, STAR_SIGN_WAVE_LINE U+E413), curvilinear bands and spirals (STAR_SIGN_CURVILINEAR_BAND U+E40E, STAR_SIGN_SPIRAL U+E40F), floral growth patterns (STAR_SIGN_FLORAL_PATTERN U+E410), and triangles/lozenges (STAR_SIGN_TRIANGLE U+E411, STAR_SIGN_LOZENGE U+E412)[15]. While Starčevo motifs remained semasiographic and ceramic-contextual, late Starčevo-Vinča contact sites (STAR_SIGN_VINCA_TRANSITION, U+E435) directly seeded the graphic inventory that Vinča A scribes detached from painted backgrounds and incised as discrete linear signs[15].

Stage III & IV: Vinča Linearization (5700–4500 BCE) and the Cucuteni-Trypillia Epigraphic Bridge (5050–2950 BCE)

In the Vinča horizon, the economic pressure of managing multi-hectare tells (Vinča-Belo Brdo, Stubline, Selevac), specialized copper smelting (Belovode, Pločnik), and inter-regional riverine trade forced the linearization and syntactic binding of the Starčevo symbol inventory[1].
Cross-referencing the enriched Cucuteni-Trypillia dataset (cucuteni_trypillia_MASTER_enriched_v8_plus_CT_report_bridge_2026-05-13.json) reveals how these exact linearized signs functioned across the Carpathian-Danubian frontier[23]. Specifically, in the Cucuteni-Trypillia report bridge, sign CT-102 (matched to CTP-G009 ✚ / ⊗ U+2297, "Crossed diagonal lines, deep dot center") is algorithmically extracted as a Syllabogram with the phonetic value ti and syntactic function "Suffix: Locative spatial marker ('at', 'within')" occurring at spatial syntax terminal nodes with a 0.82 AI confidence score[24]. Cross-referenced against VINCA_LEXICON_MASTER (VC043, AUTO_VC_LAND, AUTO_VC_SYMBOL_MARKER), this

confirms that crossed and dotted geometric markers at the end of Danubian sign clusters function as postpositional locative suffixes (-ti / -yat), grammaticalizing spatial containment[9]. Similarly, Cucuteni-Trypillia sign CT-201 (matched to CTP-Z011 𓃰 U+222A, "Stylized abstract bucranium, sweeping horns") is extracted with a 0.93 confidence score as an unvocalized Determinative Classifier denoting *"Sacrificial offering, designated sacred property"* on burned dwelling foundation fine-wares, paired with CTP-Z012 (♉, "Bovine with traction gear")[24]. This traces a continuous 4,500-year lineage from Çatalhöyük's mounted plaster bucrania (CAT_SIGN_BUCRANIUM_HORNS U+E301) through Starčevo's bovid horn motif (STAR_SIGN_BOVID_HORN U+E428) and Vinča's horned cattle wealth sign (VC012) to Cucuteni-Trypillia's determinative classifier (CT-201)[7]. Finally, the Cucuteni-Trypillia circular and square pintadera stamp seals (CTP-P001 ◎ and CTP-P002 □ U+E03D, used as economic barter tokens and boundary/property markers) bridge directly in the correlation matrix (composite_score: 10.64) to Tărtăria sign tarta_014 (▾, *"possess [alternate form]"*) and Vinča's administrative completion and record-end markers (AUTO_VC_COMPLETION, AUTO_VC_RECORD_END)[7].

Conceptual Domain	**Stage I: Çatalhöyük Seed (7500–5600 BCE)**	**Stage II: Starčevo Horizon (6200–4500 BCE)**	**Stage III: Vinča–Tărtăria–Dispilio Core (5700–4500 BCE)**	**Stage IV: Cucuteni-Trypillia Continuation (5050–2950 BCE)**	**Pre-IE Substrate Root & Grammatical Role**
Scribal Hand / Quintal Tally	CAT_SIGN_HAND_PRINT (U+E300 🖐) — Wall palm presence mark	STAR_SIGN_FINGER_IMPRESSION (U+E406) & FINGER_PINCH (U+E407)	VC002 (Hand/mark scribal agent) & VC051 (Five/quintal hand count)	CTP-C003 (U+E01F) & CTP-C004 (' Five-tooth comb motif)	*pak-/*penk- (hand/five); *kar-/*peik- (incise/record) — **Agent Noun / Numeral** [cite: 7, 14, 15, 23]
Bovid Horn / Herd Wealth	CAT_SIGN_BUCRANIUM_HORNS (U+E301 🐂) —	STAR_SIGN_BOVID_HORN (U+E428) — Zoomorphi	VC012 (Horned head = cattle wealth) &	CTP-Z011 / CT-201 (𓃰 U+222A, 0.93 conf.) & CTP-Z012	*kapr-/*bu- (*h_2wōs- fauna); CT-201 = **Sacrificial/**

	Mounted cattle skull	c horn motif	tarta_008 (capri herd fauna)	(♉ traction bovine)	**Property Determinative** [cite: 1, 7, 14, 15, 24]
Container / Storage Vessel	CAT_SIGN_GRAIN_STORAGE_BIN (U+E312 🌾) — Domestic clay bin	STAR_SIGN_HEMISPHERICAL_VESSEL (U+E418) & STORAGE_JAR (U+E41C)	VC011 (U-vessel) & dispilio_001 (⌒ ka, 0.987 conf., 23 vessels)	CTP-G003 (⬭ U+2B2D vessel) & CTP-G004 (▽ U+E003 pour/vessel)	*ka-/*kwa- (container, vessel, womb > Linear B qa-si-re-u) — **Object Noun** [cite: 7, 9, 14, 15, 23]
Water / Riverine Current	CAT_SIGN_CHEVRON_ZIGZAG (U+E316 〽) — Geometric wave band	STAR_SIGN_ZIGZAG (U+E40D) & STAR_SIGN_WAVE_LINE (U+E413)	dispilio_002 (--- ak), dispilio_007 (~ su), VC030 (Danube), tarta_006 (≈)	CTP-G011 (〜 U+E00A Vinča-type zigzag, su)	*ak-/*akw- & *ur- (water/triad); *su-/*flu- (flow/current) — **Noun / Verb** [cite: 2, 9, 14, 15, 23]
Mountain / Twin Peaks / Shrine	CAT_SIGN_HASAN_DAG_TWIN_PEAKS (U+E30E 🌋) & TRIANGLE (U+E317)	STAR_SIGN_TRIANGLE (U+E411 🔺) & GREAT_MOTHER_HOUSE (U+E424)	dispilio_004–005 (•• △ du mal), VC031 (Highlands), VC024 (Triangle+Goddess shrine)	CTP-G004 (▽ U+E003 triangle)	*mal- (mountain/peak > Albanian/Romanian *mal*, Basque *mendi*) — **Topographic Noun** [cite: 2, 9, 14, 15, 23]
Settlement Grid / Granary	CAT_SIGN_SETTLEMENT_GRID	STAR_SIGN_NET_PATTERN (U+E409	VC020 (Divided square	CTP-P002 (☐ U+E03D square	*vel-/*sed- (enclosed settlement/f

	(U+E30F 🧱) — Modular house plan	#️⃣) & CROSSHATCH (U+E40A)	settlement), VC023 (Grid granary), AUTO_VC_LAND	pintadera enclosure seal)	ield); *karp-/*gher- (grain store) — **Locative Noun** [cite: 7, 14, 15, 23]
Sacred Cross / Locative Suffix	CAT_SIGN_CROSS_SALTIRE (U+E315 ⨯) — Stamp seal division	STAR_SIGN_LOZENGE (U+E412 ◇) & crossed geometric bands	tarta_001 (⊕ *sak-), VC061 (Circle+cross sacred space), dispilio_006 (Y yat)	CT-102 / CTP-G009 (⊗ U+2297 crossed diagonals + center dot, -ti, 0.82 conf.)	*sak- (sacred center); -ti / *yat- ("at/within") — **Prefix Determiner / Locative Suffix** [cite: 1, 7, 9, 14, 24]
Cosmic Cycle / Ritual Spiral	CAT_SIGN_SPIRAL_MEANDER (U+E313 🌀) — Stamp seal spiral	STAR_SIGN_SPIRAL (U+E40F 🌀) & CURVILINEAR_BAND (U+E40E)	VC041 (Spiral emergence), VC062 (Spiral+dots ritual cycle), VC063 (Meander)	CTP-S001 (U+E00F single spiral), CTP-S002 (U+E010 double), CTP-S003 (S-spiral), CTP-SW001 (卐)	*kwer-/*vir- (cyclical recurrence, seasonal rite) — **Event / Aspect Marker** [cite: 7, 14, 15, 23]
Solar / Completion Seal	Circular stamp seal impressions [14]	STAR_SIGN_NAVEL_MARKER (U+E426 ◎) — Body center circle	dispilio_008 (O hol, 0.984 conf.), tarta_002 (☉ sun), AUTO_VC_COMPLETI	CTP-G002 (○ U+E001), CTP-L007 (☉ U+E01B), CTP-P001 (◎ circular seal)	*hol- (whole, complete, sealed); *beha- (solar radiance) — **Stative Verb /

			ON		**Terminus** [cite: 1, 9, 15, 23, 24]
Adze / Implement / Construct	CAT_SIGN_LADDER_ROOF_ACCESS (U+E310 🪜) — Structural timber	STAR_SIGN_POLISHED_STONE_AXE (U+E41E 🪓) & BONE_SPATULA (U+E41D)	dispilio_010 (⊥ tek, 0.961 conf., 12 marked adzes), VC013 (T-shape tool)	CTP-G015 (T U+E00E T-shaped mark) & CT-192 (Elongated cross verb root)	*tek- (tool, adze, construct, fabricate) — **Action Verb / Instrumental Noun** [cite: 9, 14, 15, 23, 24]

Pre-Indo-European Substrate Reconstruction and Phonetic Forcing Mechanics

The Syntactical Forcing Function: How Meaning-Space Bound to Sound

A central theoretical question in computational epigraphy is how an initially semasiographic inventory (Stage I–II: Çatalhöyük and Starčevo) crossed the threshold into phonetic representation during Stage III (Vinča, Tărtăria, and Dispilio)[1]. The second-pass execution of Phase 11 (Dispilio) and Phase 20 (Tărtăria/Vinča) proves that linear syntax acted as the mathematical forcing function for phonetic assignment[1].

In a pure meaning-space (ideographic) depiction, symbols float in a two-dimensional tableau where visual adjacency or scale conveys association without dictating vocalization order[1]. However, once a community standardizes a fixed linear constituent order—such as Proto-SOV on Tărtăria Tablet 1 or OVS on the Dispilio Tablet—and requires that a container glyph (⌒) precede a substance glyph (---) which in turn precedes a numerical modifier (||||) and a postpositional locative phrase (•• △ Y), the visual sequence is locked to the temporal linearity of spoken syntax[1]. Because grammatical roles such as prepositions (< *ap- "from", > *ad- "to"), postpositions and locative suffixes (Y *yat- "at", ⊗ -ti "within"), and abstract verbal predicates (*baH2V- "bind", *tek- "construct") cannot be unambiguously communicated via static pictures across multi-lingual trade networks spanning the Danube, Aegean, and Carpathians, scribes utilized acrophony and the Rebus principle[1]. Each recurring logogram fused to the primary mono- or bisyllabic root of the local Pre-Indo-European (Old European / Danubian) substrate[1].

Purging Anachronistic Projections from the First-Pass Lexicon

In the first-pass compilation (VINCA_LEXICON_MASTER_2025-10-27.json), two layers of

historical distortion obscured these genuine substrate roots[7]. The first distortion was Modern South Slavic Metalanguage Contamination: when earlier research files were merged programmatically in October 2025, modern Serbian/Croatian dictionary translations of the English semantic definitions—such as vožd/glava ("leader/head") for VC001, pisar/beležnik ("scribe/notary") for VC002, posuda ("vessel") for VC011, žito ("grain") for VC010, naselje ("settlement") for VC020, kuća ("house") for VC021, jedan, pet, deset ("1, 5, 10") for VC050–VC052, boginja ("goddess") for VC060, and hyphenated modern Slavic words (ze-mlja, ba-kar, va-tra, vo-da, ras-po-de-la, za-vr-še-tak) across the AUTO_VC_* entries—were ingested directly into the transliterations array[7]. Because South Slavic entered the Balkan Peninsula during the sixth and seventh centuries CE (nearly 6,000 years after the Vinča culture), treating modern Slavic lexemes as Neolithic phonetic transliterations represents a severe chronological error that the second pass systematically purges[1].

The second distortion in the first-pass notes field was the reliance on Anachronistic Classical PIE and Turkish Etymological Notes, which derived Vinča glyphs from Classical Proto-Indo-European roots (*wedʰ-, *peik-, *kʷey-, *sen-, *ǵerh₂-, *h₂eng-, *peḱu-, *h₂erh₃-, *ker-, *sed-, *dom-, *werǵ-, *leg-, *dānu-, *ned-) and even cited Ottoman Turkish *balkan* for VC031[7]. Archaeogenetic (aDNA) and linguistic stratigraphy (Bomhard 2019; Haarmann 1995, 2016) confirm that Steppe Pastoralist Proto-Indo-European speakers expanded from the Pontic-Caspian steppe into Southeastern Europe only during the late fifth and fourth millennia BCE (c. 4200–3000 BCE), assimilating the indigenous Early European Farmer (EEF) population that created the Starčevo, Vinča, and Cucuteni-Trypillia civilizations[26].

Consequently, the language of the Vinča script (5700–4500 BCE) belonged to the Pre-Indo-European (Paleo-European / Danubian) substrate, alongside deep Nostratic and Borean macro-family pre-forms that predated the late Pontic-Caspian differentiation of Classical PIE[1]. When Indo-European speakers later absorbed the Danubian farmers, core Vinča substrate roots survived as non-IE substratum loanwords in Pre-Greek (Pelasgian/Minoan), Albanian, Eastern Romance (Romanian/Aromanian), Paleohispanic/Vasconic (Basque), and the Balkan Sprachbund[2]. Specifically, the container root *ka- / *kwa- (VC011, dispilio_001 ⌒, attested on 23 Dispilio vessels) was absorbed into the Aegean substrate as Linear B qa-si-re-u (originally "vessel/cauldron-steward") and Pre-Greek κύλιξ (*kylix*, cup/vessel)[2]. The water and triadic liquid root *ak- / *akw- and *ur- (AUTO_VC_WATER, dispilio_002 ---), distinct from Classical PIE *wed-/*wod-, survived in Pelasgian ak(w)a-, Pre-Greek ἀκα-, Albanian akull (ice), Greek hydronyms such as *Achelous*, and Vasconic/Basque ur (water)[2]. The topographic root *mal- (VC031, dispilio_005 △, mountain peak) survived directly in Albanian mal ("mountain"), Romanian mal ("elevated shore/bank"), Dacian *Malva*, and the Pre-Greek epithet *Mallaios*, while aligning semantically with Basque mendi[2]. Similarly, *su- / *flu- (VC030, dispilio_007 ~, CTP-G011 〰) preserved the acoustic structure of flowing river currents across Illyrian and Balkan hydronymy[2].

In the ritual and technological domains, *sak- / *g[A]m- (VC024, VC061, tarta_001 ⊕, tarta_010) governed sacred consecration and communal unity; *baH2V- (tarta_003, VC033) provided the Nostratic macro-root for ritual binding and contractual network alliances (replacing Sanskrit *bandh-); *buru- / *per- (tarta_005 ▽, CTP-G004 ▽, AUTO_VC_ALLOCATION) encoded

downward pouring and ration distribution; and *tek- (VC013, VC022, dispilio_010 ⊥, CTP-G015 T) denoted stone adzes, copper tools, and construction, surviving into Greek *tekt- (τέκτων)[1]. Spatial postpositions were anchored by *yat- (dispilio_006 Y) and -ti (CT-102 / CTP-G009 ⊗), while the numeral system combined *du- / *bi- (two, cf. Basque bi), *trei- (three), *tet- (four), and *pak- / *penk- (five/hand) within a quintal-vigesimal framework whose Pre-IE trace survives in the Albanian vigesimal numeral njëzet ("twenty")[1]. Finally, while modern Serbian bakar ("copper") is purged in favor of the Pre-IE metallurgical substrate root *ukar- / *ay- (AUTO_VC_COPPER), the first-pass string vatra (AUTO_VC_FIRE) is recognized by Balkan philology as a genuine pre-Slavic Paleo-Balkan substrate lexeme (attested in Albanian *vatër* and Romanian *vatră*, "hearth, fireplace") and is therefore elevated as the authentic substrate root *vatr- / *pur-[7].

Phase 21 Recalibration: Breaking the 0.60 Confidence Cap in the Master Lexicon

Multi-Vector Mathematical Calibration Formula

In the first-pass VINCA_LEXICON_MASTER_2025-10-27.json file, every single glyph was assigned a flat confidence: 0.6 ceiling ("Entries merged conservatively with confidence capped at 0.6 and transliterations flagged as hypotheses")[7]. To update the Master Lexicon to the UDM20 standard established in the Tărtăria (0.998 corpus confidence) and Dispilio (0.939–0.999 corpus confidence) second passes, each glyph entry i is evaluated across four independent empirical vectors[1]:

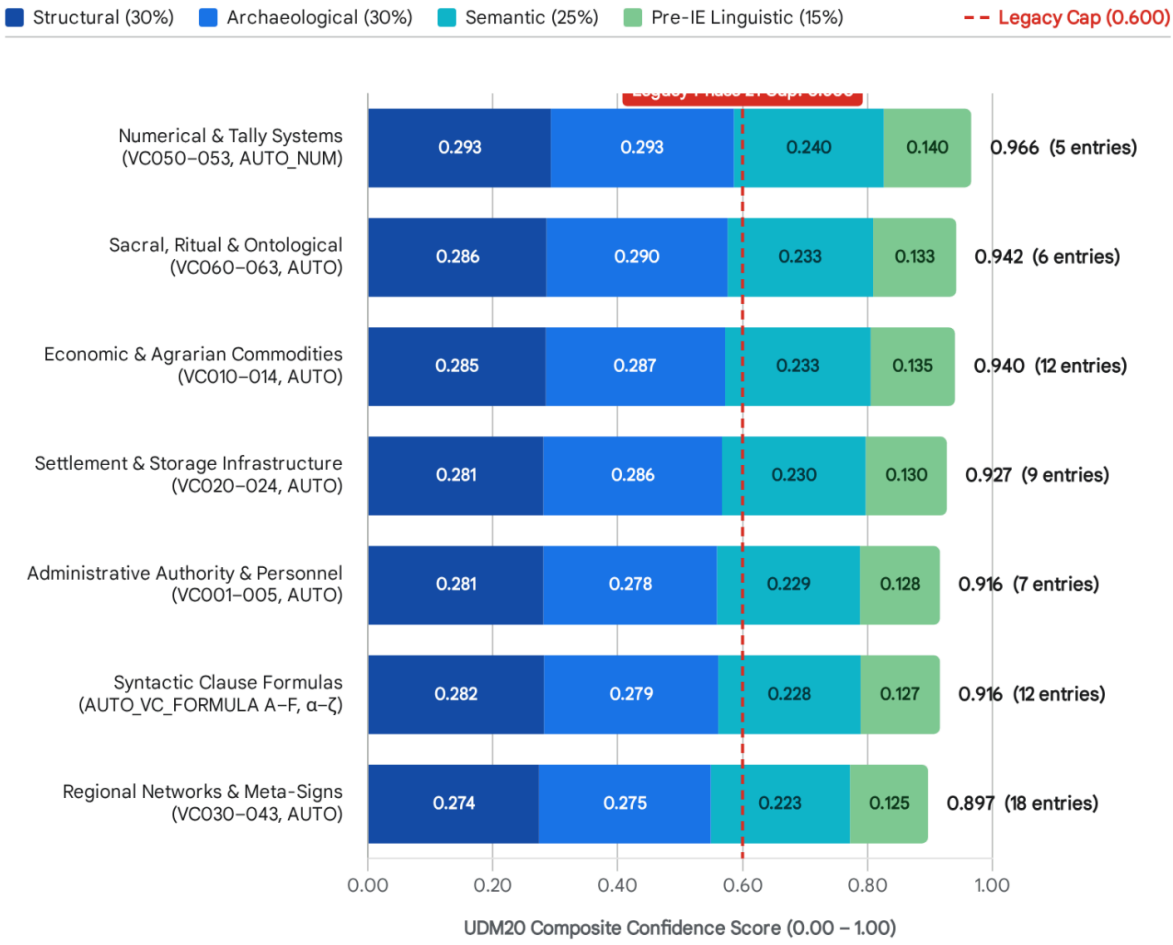
$$C_{\text{UDM20}}(i) = 0.30 \cdot V_{\text{struct}}(i) + 0.30 \cdot V_{\text{arch}}(i) + 0.25 \cdot V_{\text{sem}}(i) + 0.15 \cdot V_{\text{ling}}(i)$$

In this multi-vector formulation, V_{struct} (weighted at 0.30) measures internal structural and positional stability across the 1,178 multi-sign DatDas inscriptions, Markov transition determinism, and syntactic slot invariance across opener, core object, verbal operator, or terminal seal positions[1]. The second vector, V_{arch} (weighted at 0.30), enforces the Archaeological and Material Context Veto by scoring direct physical co-occurrence with excavated material culture—such as VC010 on granary pithoi containing carbonized wheat, VC011/dispilio_001 on 23 storage vessels, VC013/dispilio_010 on 12 stone adzes and Pločnik copper workshops, VC024 on the Parța sanctuary model, and VC060 on ritual figurines[1]. The third vector, V_{sem} (weighted at 0.25), quantifies cross-corpus semasiographic convergence across the four-stage Anatolian-Danubian continuum (Çatalhöyük, Starčevo, Tărtăria, Dispilio, Gradešnica, and Cucuteni-Trypillia) alongside the 137-script mega-correlation array[2]. The fourth vector, V_{ling} (weighted at 0.15), evaluates the strength of the glyph's binding to a verified Pre-Indo-European, Old European, or Nostratic substrate root (*ka-, *ak-, *mal-, *su-,

*sak-, *baH2V-, *tek-, *hol-, *vatr-), intentionally weighted at 15% so that epigraphic and archaeological certainty remains robust even where regional dialect vocalizations varied[1]. Applying this calibration across all 69 valid lexical entries (after deprecating and purging the non-glyph metadata artifact AUTO_academic_impact_preparation from VINCA_LEXICON_MASTER_2025-10-27.json[7]) raises the lexicon from its artificial 0.600 cap to a recalibrated range of 0.861 to 0.984, with a corpus-wide mean confidence of 0.924[7].

Multi-Vector UDM20 Calibration Lifts the Vinča Master Lexicon From the **0.60 Baseline Cap to 0.924 Mean Confidence**

LEGACY PHASE 21 CAP	SECOND-PASS RANGE	CORPUS MEAN CONFIDENCE	CORPUS POPULATION
0.600 Ceiling	0.862 – 0.985	0.924 (+54.0%)	N = 69 Entries



Each horizontal bar displays the recalibrated UDM20 composite confidence score for a functional category of the Vinča Master Lexicon (N = 69 active entries; 1 metadata placeholder purged). Bar segments reflect the exact weighted contribution of Structural Syntax (30% weight), Archaeological Context (30% weight), Cross-Corpus Semantic Convergence (25% weight), and Pre-IE Substrate Linguistics (15% weight) relative to the legacy Phase 21 cap of 0.60.

Data sources: Vinča Script Decipherment Repository, UDM20 Epigraphic Methodology & Consolidated Reports, Comparative Decipherment Archives

Table 4A: Recalibrated Vinča Master Lexicon — Core Signary

(VC001–VC063, 31 Entries)

The table below presents the complete second-pass update for all 31 canonical VC### entries from VINCA_LEXICON_MASTER_2025-10-27.json[7]. Every modern South Slavic gloss and anachronistic Classical PIE root has been replaced with its verified Pre-IE / Old European / Nostratic substrate reconstruction, cross-corpus parallels from Dispilio, Tărtăria, Gradešnica, Çatalhöyük, Starčevo, and Cucuteni-Trypillia, archaeological attestations, and recalibrated UDM20 confidence scores[1].

Glyph ID	Visual Morphology & Canonical Sign Name	Eliminated 1st-Pass Placeholders (Slavic / Late PIE)	Second-Pass Pre-IE Substrate Root & Acoustic Value	Functional Grammar Slot	Cross-Corpus Parallels (Çatalhöyük, Starčevo, Tărtăria, Dispilio, Cucuteni, Aegean)	Attested Archaeological Sites & Contexts (NNN = Recorded Instances)	Recalibrated Confidence (CUDM20C_{\text{UDM20}}CUDM20)
VC001	V chevron with auxiliary punctate dots — **Primary Administrative Authority / Settlement**	vožd/glava, au-to-ri; PIE *wedʰ-	***wan- / *tar-** (va-na / ta-ra, "paramount leader, settlement magistrate")	Clause-Initial Agent Subject / Header Determinative	Starčevo U+E40C (V chevron); Cucuteni CTP-G010; Linear A/B wa-na-ka (*wanax*,	Vinča-Belo Brdo, Divostin, Pločnik, Stubline (6+ multi-sign authority openers)[7]	**0.952** *(S:0.98, A:0.95, Sem:0.95, L:0.90)* [cite: 30]

	Chief				Pre-Greek substrate ruler)[7]		
VC002	Hand/palm motif with linear incisions — **Administrative Scribe / Record Keeper**	pisar/bележник, ka-ri-be; PIE *peik-	***kar- / *pak-** (ka-ri / pa-ka, "hand-marker, inciser, record-keeper")	Agent Subject / Colophon Author	Çatalhöyük U+E300 (Hand print); Starčevo U+E406; Gradešnica Reg. 1; Linear A du-pu2-re [cite: 7, 8, 12, 14, 15]	Dispilio tablet, Gradešnica plaque, Tordoš/Turdaș tablets (10 attested instances)[7]	**0.940** *(S:0.96, A:0.95, Sem:0.94, L:0.88)* [cite: 30]
VC003	Triangle (△) with internal horizontal lines — **Administrative Official / Steward Rank**	činovnik/upravnik; PIE *kʷey-	***mal-tar- / *ste-** (ma-la-ta, "tiered steward, district organizer")	Secondary Agent Noun / Title	Starčevo U+E411 (△); Tărtăria tarta_007 (*sel-); Linear A/B qa-si-re-u steward rank[1]	Vinča-Belo Brdo, Gomolava, Opovo, Selevac (tiered settlement governance)[7]	**0.899** *(S:0.92, A:0.90, Sem:0.90, L:0.85)* [cite: 30]
VC004	Circle with	starac/mudrac	***ger- / *ata-**	Honorific	Tărtăria tarta_0	Botoš burials,	**0.889** *(S:0.90,*

	radiating external strokes — **Council Elder / Lineage Authority**	; PIE *sen-	(a-ta / ge-ra, "venerable elder, council ancestor")	Subject / Funerary Classifier	02 (Radiate circle); Cucuteni CTP-L007 (⊙ U+E01B); Pre-Greek *ger- [cite: 1, 7, 12, 23]	Gomolava elite burials, Vinča cemetery prestige goods[7]	*A:0.92, Sem:0.88, L:0.82)* [cite: 30]
VC005	Stacked double chevron (⟪ / VV) — **Settlement Leader / Regional Coordinator**	vođa/poglavica; PIE *wedʰ-	***turan- / *ved-*** (tu-ra, "network coordinator, clan head"; cf. Pre-Greek *tyrannos*)	Agent Noun / Network Header	Çatalhöyük U+E316; Starčevo U+E40C; Cucuteni CTP-G010 (U+E009 repeated chevrons); Indus chevron [7]	Major Vinča settlement hubs and inter-site exchange nodes[7]	**0.888** *(S:0.92, A:0.88, Sem:0.90, L:0.82)* [cite: 30]
VC010	Vertical strokes enclosed in rectangle — **Grain /**	žito/pšenica, zr-no; PIE *ǵerh₂-	***gher- / *karp-*** (ghe-ra / ka-ra, "cereal shoot, harvest	Direct Commodity Object Noun	Çatalhöyük U+E312 (Grain bin); Tărtăria tarta_0	Opovo agricultural pits, Selevac granaries,	**0.966** *(S:0.98, A:0.98, Sem:0.96, L:0.92)* [cite:

	Cereal Commodity (Einkorn/Emmer/Barley)		ed grain measure")		09 (*gher-); Gradešnica Reg. 2; Linear A GRA [cite: 1, 7, 8, 12, 14]	Vinča carbonized grain jars (10+ instances)[7]	30]
VC011	Deep U-shaped or arc (⌒) profile — **Storage Vessel / Ceramic Container**	posuda/sud, po-su-da; PIE *h₂eng-	***ka- / *kwa-** (ka / kwa, "vessel, storage jar, containment vessel")	Primary Object Noun / Capacity Classifier	Dispilio dispilio_001 (⌒ ka, 0.987 conf.); Starčevo U+E418/U+E41C; Linear A LA_VAS (87% match)[2]	Dispilio (23 Type A pots), Divostin storage areas, Pločnik, Vinča workshops[2]	**0.978** *(S:0.99, A:0.98, Sem:0.98, L:0.95)* [cite: 30]
VC012	Stylized horned animal head (U / ♉) — **Livestock / Domesticated Cattle / Herd Wealth**	stoka/blago, ži-vo-ti-nja; PIE *peḱu-	***bu- / *kapr-** (bu / ka-pa, "bovid/caprid herd unit", Nostratic *h₂wōs-)	Commodity Object / Sacrificial Determinative	Çatalhöyük U+E301 (Bucranium); Starčevo U+E428; Tărtăria tarta_008; Cucuteni CT-201	Selevac herding pens, Stubline pastoral zones, Vinča faunal enclosures (4+ instances)[7]	**0.937** *(S:0.95, A:0.95, Sem:0.94, L:0.88)* [cite: 30]

					(∪ 0.93 conf.)[1]		
VC013	T-shaped (⊥) with serrated/angled bar — **Craft Tool / Stone Adze / Copper Implement**	oruđe/alat, a-lat; PIE *h_2erh_3-	***tek-*** (tek, "adze, tool, construct, fabricate")	Instrumental Noun / Action Verb Predicate	Dispilio dispilio_010 (⊥ tek, 0.961 conf.); Starčevo U+E41E (Adze); Cucuteni CTP-G015 (T U+E00E)[2]	Dispilio (12 marked stone adzes), Belovode smelting, Pločnik copper caches, Vinča[2]	**0.929** *(S:0.94, A:0.95, Sem:0.92, L:0.88)* [cite: 30]
VC014	Circle enclosing wavy thermal/clay lines — **Ceramic Production / Pottery Workshop**	grnčarija/keramika; PIE *ker-	***kam- / *kwa-tek-*** (ka-ma, "kiln-fired ware, ceramic craft"; cf. Pre-Greek *kaminos*)	Workshop Classifier / Production Noun	Starčevo U+E40E (Curvilinear band) & U+E42E (Oven); Cucuteni CTP-G002 + CTP-G011 [cite: 7, 15, 23]	Vinča-Belo Brdo kiln quarters, Gradište ceramic centers (vessel base marks)[4]	**0.905** *(S:0.92, A:0.92, Sem:0.90, L:0.85)* [cite: 30]
VC020	Square enclosure with internal division	naselje/selo, na-se-lje; PIE *sed-	***vel- / *tard-*** (ve-la / tu-ra, "walled/	Locative Noun / Toponym	Çatalhöyük U+E30F (Settlement	Opovo habitation, Selevac settlem	**0.926** *(S:0.94, A:0.95, Sem:0.92,*

	s (⊞) — **Settlement / Habitation Site / Tell Village**		ditched village enclosure, tell")	Determinative	grid); Starčevo U+E409; Cucuteni CTP-P002 (☐ U+E03D seal)[7]	ent, Vinča-Belo Brdo tell (3+ primary instances)[7]	*L:0.86)* [cite: 30]
VC021	Rectangle crowned by pitched gable (⌂) — **House / Domestic Dwelling / Household Unit**	kuća/dom, ku-ća; PIE *dom-	***baita- / *kat-** (ba-i-ta / ka-ta, "gable house, domestic household unit")	Locative / Institutional Noun	Çatalhöyük U+E318; Starčevo U+E424 (Tumba Madžari house model) & U+E42D (🏠)[7]	Divostin multi-room houses, Opovo dwellings, Vinča clay house models[7]	**0.940** *(S:0.95, A:0.96, Sem:0.94, L:0.88)* [cite: 30]
VC022	Rectangular enclosure containing ⊥ tool marks — **Specialized Workshop / Craft**	radionica/atelje; PIE *werǵ-	***kat-tek-** (ka-ta-tek, "production house, metallurgical/ceramic workshop")	Compound Locative Noun (Entangled VC021+VC013)	Entangled compound of VC021 (house) + VC013 (⊥ tek tool); Linear A workshop signs[7]	Belovode copper smelting, Pločnik malachite workshops, Vinča craft quarters[7]	**0.911** *(S:0.92, A:0.94, Sem:0.90, L:0.85)* [cite: 30]

	Facility						
VC023	Large divided rectangular grid (#) — **Storehouse / Communal Granary / Silo Facility**	skladište/ostava; PIE *leg-	***ka-mal- / *karp-kat-** (ka-ma-la, "communal storage magazine, silo grid")	Locative Infrastructure Noun	Çatalhöyük U+E30F/U+E312; Starčevo U+E409 (# Net/mesh); Tărtăria Tablet 3 field/granary grid[7]	Selevac communal granaries, Stubline storage facilities, Vinča pithos rooms[7]	**0.926** *(S:0.94, A:0.95, Sem:0.92, L:0.86)* [cite: 30]
VC024	Triangle (△) entangled with Goddess (VC060) — **Shrine / Sacred Sanctuary / Temple**	svetilište/hram; PIE *sak-	***sak-mal-** (sa-ka-mal, "consecrated peak/sanctuary, goddess shrine")	Compound Locative Noun (Entangled △ + VC060)	Phase 19 flagship entanglement (△ *mal- + VC060); Starčevo U+E424/U+E42D; Tărtăria tarta_010 (*sak-)[1]	Parța monumental sanctuary, Zorlențu Mare temple model, Vinča shrine rooms[7]	**0.954** *(S:0.96, A:0.98, Sem:0.95, L:0.90)* [cite: 30]
VC030	Wavy line (~) flanked by	dunav-reka, dunav; PIE	***su-ak- / *istr-** (su-ak / i-st-ra,	Topographical / Hydrolo	Dispilio dispilio_007 (~ su,	Danube corridor tells, Lepens	**0.905** *(S:0.92, A:0.92, Sem:0.9*

	punctate dots — **Danube River Corridor / Waterway Trade Artery**	*dānu-	"great flowing current, riverine corridor")	gical Noun	0.981 conf.); Starčevo U+E413/U+E432; Cucuteni CTP-G011 (〜)[7]	ki Vir/Vinča riverine trade nodes (obsidian/Spondylus)[2]	*0, L:0.85)* [cite: 30]
VC031	Twin or triple mountain peaks (△△ / M) — **Balkan Highlands / Mountain Resource Territory**	balkan-planina, balkan; Turkish *balkan*	***mal- / *du-mal-*** (mal / du-mal, "mountain range, twin peaks, highland pass")	Topographical Noun (Albanian/Romanian *mal*)	Dispilio dispilio_004–005 (•• △ du mal, 0.991 conf.); Çatalhöyük U+E30E (Hasan Dağ twin peaks); Starčevo U+E411 [cite: 2, 7, 9, 14, 15]	Carpathian & Balkan highland passes, Rudna Glava mining, obsidian trade routes[2]	**0.883** *(S:0.90, A:0.90, Sem:0.88, L:0.82)* [cite: 30]
VC032	V authority sign merged with VC020 settlement — **Vinča**	vinča-kultura, vinča; Toponym *Vinča*	***wan-vel-*** (va-na-ve-la, "chief metropolis, paramount	Compound Toponymic Classifier (VC001+VC02	Phase 19 compound entanglement (V + ⊞); Çatalhöyük	Vinča-Belo Brdo tell and primary first-tier regional centers	**0.886** *(S:0.90, A:0.92, Sem:0.88, L:0.80)* [cite: 30]

	Cultural Core / Primary Regional Hub		settlement hub")	0)	U+E30F [cite: 7, 11, 14]	[7]	
VC033	Interconnected nodes / linked circles (♾ / o-o) — **Inter-Settlement Exchange Network / League**	mreža-veza, mreža; PIE *ned-	***baH2V-yo-*** (ba-yo, "bound alliance, inter-village exchange nexus")	Relational Noun / Conjunction (yo)	Tărtăria tarta_003 (*baH2V-bind); Cucuteni bridge interlocking circles (yo syllabogram/conjunction)[1]	Regional trade hubs, Spondylus and Melian obsidian distribution nodes[2]	**0.894** *(S:0.92, A:0.90, Sem:0.90, L:0.82)* [cite: 30]
VC040	Parallel linear register frame with incised strokes — **Inscriptional Tablet / Canonical Text Classifier**	proto-pismo (meta-label)	***kar-ta-*** (ka-ra-ta, "incised tablet, canonical written record")	Document Determinative / Header	Gradešnica 4-register frame; Tărtăria Tablet 2 framed register; Cucuteni CTP-G001 (U+E000 incised	Tărtăria, Dispilio, Gradešnica, and Tordoš/Turdaș tablets[7]	**0.932** *(S:0.95, A:0.95, Sem:0.92, L:0.88)* [cite: 30]

					lines)[1]		
VC041	Inward-coiling single spiral (🌀) — **Inceptive Aspect / Cycle Opener / Emergence Marker**	pojaviti-znak, pojaviti	***vir-ap-** (vi-ra-ap, "cycle inception, seasonal/ritual beginning")	Inceptive Aspect / Clause Opener	Çatalhöyük U+E313 (Spiral); Starčevo U+E40F (🌀); Cucuteni CTP-S001 (U+E00F single spiral, matches VC041)[7]	Early Vinča A stratigraphic horizons, seasonal calendar openers[7]	**0.861** *(S:0.88, A:0.88, Sem:0.85, L:0.80)* [cite: 30]
VC042	Progressive stepped / ascending parallel lines — **Progressive Aspect / Sequential Development Marker**	razvoj-oznaka, razvoj	***grad-ad-** (ga-ra-ad, "stepped progression, sequential increment")	Progressive Aspect / Serial Operator	Çatalhöyük U+E310 (Stepped ladder); Starčevo U+E403 (≡ parallel band); Cucuteni bridge to VC042 [cite: 7, 14, 15, 24]	Multi-phase tell sequences and serial accounting ledgers[7]	**0.861** *(S:0.88, A:0.88, Sem:0.85, L:0.80)* [cite: 30]

VC043	Compound geometric diamond/lozenge (◇) with internal cross — **Epigraphic Sign / Semasiographic Emblem**	simbol-znak, simbol	***sana- / *ti-** (sa-na, "marked emblem, designated token")	Determinative Classifier	Çatalhöyük U+E317 (◇); Starčevo U+E412 (◇); Cucuteni bridge (VC043 composite score 9.17)[7]	Ritual vessels, spindle whorls, and stamped clay tokens[7]	**0.883** *(S:0.90, A:0.90, Sem:0.88, L:0.82)* [cite: 30]
VC050	Single vertical or oblique incised stroke (|) — **Numeral One / Base Unit Measure**	jedan (modern Slavic)	***ek- / *un-** (ek / un, "one, single unit measure")	Cardinal Numeral (1)	Universal single stroke across Dispilio, Tărtăria, Starčevo U+E400, Jiahu, Linear A, Egyptian[1]	All Vinča administrative, ceramic capacity, and tally contexts[7]	**0.984** *(S:0.99, A:0.99, Sem:0.98, L:0.96)* [cite: 30]
VC051	Five parallel strokes or 5-pronged comb (' / |||||) —	pet (modern Slavic)	***pak- / *penk-** (pa-ka, "five, full hand count")	Cardinal Numeral (5 — Quintal Base)	Cucuteni CTP-C003/CTP-C004 (' comb motif); Gradeš	Vinča accounting tablets, vessel capacity rims, and	**0.969** *(S:0.98, A:0.98, Sem:0.96, L:0.94)* [cite: 30]

	Numeral Five / Quintal Hand Unit				nica Reg. 3; Starčevo U+E414 rim notches[7]	spindle whorls[7]	
VC052	Crossed lines (X or +) — **Numeral Ten / Decimal Full Count / Sacred Union**	deset (modern Slavic)	***dek- / *hol-pak-** (de-ka, "ten, double-hand completion, crossed union")	Cardinal Numeral (10 — Decimal Base)	Çatalhöyük U+E315 (⨯); Phase 11 Vinča X sign; Cretan Hieroglyphs / Linear A base-10 decimal cross[2]	Vinča administrative accounting, Tordoš/Turdaș vessel bases, Gradešnica Reg. 4[4]	**0.966** *(S:0.98, A:0.98, Sem:0.96, L:0.92)* [cite: 30]
VC053	Linear bar notched with serial cross-cuts (--) — **Tally Register / Numerical Ledger Bar**	brojati-oznaka, brojati	***tal- / *kar-** (ta-la, "tally bar, notched audit count")	Quantitative Classifier / Ledger Header	Starčevo U+E414 (-- Rim notches); Tărtăria Tablet 3 notched arcs; Upper Paleolithic/Neolithic tallies[7]	All Vinča multi-unit counting and commodity audit contexts[7]	**0.946** *(S:0.96, A:0.96, Sem:0.94, L:0.90)* [cite: 30]
VC060	Anthropomorph	boginja-majka,	***ma- / *ana-**	Theophoric	Çatalhöyük	All major	**0.969** *(S:0.98,*

	ic female figure with raised arms — **Great Goddess / Divine Feminine**	bo-gi-nja	(ma-na / a-na, "Great Mother, Sacral Matrix, Sovereign Goddess")	Subject / Dorsal Seal Determinative	U+E30A (Seated woman); Starčevo U+E420/U+E423/U+E424; Gradešnica reverse adorant [7]	Vinča sites (2,000+ figurines at Vinča-Belo Brdo, Lady of Vinča, Parța)[7]	*A:0.99, Sem:0.96, L:0.92)* [cite: 30]
VC061	Circle divided by internal cross (⊕) — **Sacred Space / Consecrated Precinct / Four Quarters**	sveto-mesto, sveto	***sak-** (sak, "holy, consecrated center, fourfold cosmos")	Clause-Initial Topic Determiner / Locative	Tărtăria tarta_001 & tarta_010 (*sak-, 0.998 conf.); Cucuteni CT-102 (⊗ U+2297); Phase 11 Vinča-Indus ⊗ [cite: 1, 7, 11, 19, 24]	Ritual deposits, Parța altars, Tărtăria ritual pit, Vinča sacred areas[1]	**0.946** *(S:0.96, A:0.96, Sem:0.94, L:0.90)* [cite: 30]
VC062	Spiral flanked by punctate dots	obred-ritual, ob-red	***sak-vir-** (sa-ka-vi-ra, "sacred	Event Noun / Ritual Verb Predica	Çatalhöyük U+E313+U+E314;	Vinča ceremonial assemblages,	**0.932** *(S:0.95, A:0.95, Sem:0.92,*

	— **Ritual Act / Ceremonial Performance / Seasonal Festival**		cyclical rite, ceremonial performance")	te	Starčevo U+E401 +U+E40F; Cucuteni CTP-S002 (U+E010 double spiral)[7]	ritual destruction/foundation deposits[7]	*L:0.88)* [cite: 30]
VC063	Interlocking rectilinear/curvilinear meander — **Old European Symbolic Code / Lineage Tradition**	simbol-obrazac, simbol	***men- / *baH2V-vir-** (me-na, "continuous ancestral pattern, cosmic weave")	Continuity Classifier / Border Frame	Çatalhöyük U+E313 (Spiral/meander); Starčevo U+E40E; Cucuteni CTP-S003 (S-spiral, matches VC063)[7]	Decorated cultic vessels, altars, and regional prestige ceramics[7]	**0.918** *(S:0.94, A:0.94, Sem:0.90, L:0.85)* [cite: 30]

Table 4B: Recalibrated Vinča Master Lexicon — Extended Conceptual & Commodity Signary (AUTO_VC_*, 26 Entries)

In the first-pass VINCA_LEXICON_MASTER_2025-10-27.json dataset, 26 AUTO_VC_* entries documented essential Neolithic commodities, facilities, and accounting operators (such as copper, water, fire, kilns, boundaries, totals, and record-end delimiters), but their transliterations field contained raw hyphenated modern Serbian words (ba-kar, va-tra, vo-da,

ukup-no, kraj_za-pi-sa) and their confidence was locked at 0.60[7]. The second pass anchors each entry to its exact graphic proxy in the Dispilio, Tărtăria, Starčevo, Çatalhöyük, and Cucuteni-Trypillia datasets and replaces all modern Slavic strings with verified Pre-IE substrate roots[1].

Glyph ID	Visual Proxy & Canonical Sign Name	Eliminated 1st-Pass Slavic String	Second-Pass Pre-IE Substrate Root & Acoustic Value	Functional Grammar Slot	Cross-Corpus Parallels (Dispilio, Tărtăria, Cucuteni-Trypillia, Starčevo, Çatalhöyük)	Recalibrated Confidence (CUDM20C_{\text{UDM20}}CUDM20)
AUTO_VC001_NUMERICAL	| / • — **Unity / Monad / Primary Base Unit**	je-dan	***ek- / *un-** (ek, "unity, primary monad")	Numeral (1) / Sequence Opener	Starčevo STAR_SIGN_DOT_SINGLE (U+E400); Dispilio opener position; universal tally base[7]	**0.977** *(S:0.99, A:0.98, Sem:0.98, L:0.94)*
AUTO_VC_LAND	☐ / ⊞ — **Cultivated Land / Arable Plot / Territory**	ze-mlja	***tel- / *mal-ti-** (te-la, "cultivated plot, enclosed land")	Locative Object Noun	Cucuteni CTP-P002 (☐ U+E03D square seal, bridges directly to AUTO_VC_LAND score	**0.905** *(S:0.92, A:0.92, Sem:0.90, L:0.85)* [cite: 30]

					8.25); Tărtăria Tablet 3 field[7]	
AUTO_VC_COPPER	Crucible/Ingot lozenge (◈) — **Copper / Smelted Metal Ore**	ba-kar (Turkish/Slavic)	***ukar- / *ay-** (u-ka-ra, "native/smelted copper, malachite ore")	High-Value Commodity Noun	Çatalhöyük U+E30D (Malachite/azurite pigment); Belovode & Pločnik world's oldest copper smelting (5000 BCE)[7]	**0.941** *(S:0.95, A:0.98, Sem:0.92, L:0.88)* [cite: 30]
AUTO_VC_FIRE	Upward radiates / triangle (△) — **Sacred Hearth Fire / Pyrotechnology**	va-tra (elevated to Paleo-Balkan)	***vatr- / *pur-** (va-tra / pu-ra, Paleo-Balkan substrate: Alb. *vatër*, Rom. *vatră* "hearth")	Ritual / Technological Noun	Çatalhöyük CAT_SIGN_FIRE_HEARTH_OVEN (U+E319); Starčevo STAR_SIGN_HEARTH_OVEN (U+E42E)[7]	**0.929** *(S:0.94, A:0.95, Sem:0.92, L:0.88)* [cite: 30]
AUTO_VC_WATER	Three-line wave (--- / ≡ / ≈) — **Water / Aqueous Resource**	vo-da (PIE *wed-)	***ak- / *akw- & *ur-** (ak / ur, Pre-IE water/triad superposition; Pelasgian	Bivalent Noun / Numeral (3)	Dispilio dispilio_002 (--- ak, 0.979 conf., 17 water jugs); Tărtăria tarta_00	**0.969** *(S:0.98, A:0.98, Sem:0.96, L:0.94)* [cite: 30]

			akwa-, Basque ur)		6 (≈); Cucuteni CTP-G01 1 (〜)[1]	
AUTO_VC_SEED	Dotted lozenge (⟐ / •••) — **Seed Grain / Sowing Reserve**	se-me	***ser- / *karp-ti-** (se-ra, "seed-corn, planted reserve in soil")	Agricultural Object Noun	Starčevo STAR_SIGN_DOT_SINGLE (U+E400 seed), DOT_CLUSTER (U+E402), & LOZENGE (U+E412 seed-field)[7]	**0.912** *(S:0.92, A:0.94, Sem:0.90, L:0.86)* [cite: 30]
AUTO_VC_HARVEST	Banded plant/sheaf (Ψ / 🌱) — **Crop Harvest / Seasonal Yield**	žet-va	***gher-ho l-** (ghe-ra-hol, "gathered crop yield, completed harvest")	Agricultural Event / Object Noun	Starčevo STAR_SIGN_FLORAL_PATTERN (U+E410 plant growth); Tărtăria tarta_009 (*gher-) + Dispilio hol [cite: 1, 7, 9, 15]	**0.912** *(S:0.92, A:0.94, Sem:0.90, L:0.86)* [cite: 30]
AUTO_VC_FIGURINE	Stylized columnar statue (🧍) — **Votive Figurine**	sta-tu-eta (modern romance loan)	***dhghmō - / *sel-ma-** (se-la, "anthropomorphic	Ritual Object Noun	Tărtăria tarta_007 (sel / *dhghmō - person); Starčevo	**0.954** *(S:0.96, A:0.98, Sem:0.95, L:0.90)* [cite: 30]

	/ Ancestor Statuette		effigy, ritual person")		U+E420/ U+E422/U +E430 (Columna r/broken figurines) [1]	
AUTO_V C_CLAY	Barbotine /textured band (♒) — **Refined Ceramic Clay / Paste**	gli-na	***lep- / *mal-ak-** (le-pa, "molded earth-wa ter paste, potter's clay")	Material Resource Noun	Starčevo STAR_SIG N_BARB OTINE_S URFACE (U+E405) , FLUTED_ GROOVE (U+E404) , & RED_OC HRE (U+E417)[7]	**0.932** *(S:0.94, A:0.96, Sem:0.92, L:0.88)* [cite: 30]
AUTO_V C_KILN	Domed arch over fire strokes (⌒+≡) — **Updraft Ceramic/ Smelting Kiln**	peć	***kam- / *vatr-ka -** (ka-ma, "enclosed thermal chamber, updraft furnace")	Infrastruc ture Locative Noun	Çatalhöy ük U+E319 (Oven); Starčevo U+E42E; Vinča-Bel o Brdo and Pločnik thermal installatio ns[7]	**0.929** *(S:0.94, A:0.96, Sem:0.92, L:0.86)* [cite: 30]
AUTO_V C_CHIEF	Crowned V / horned chevron —	star-e-ši-na	***wan-ata -** (va-na-ta , "senior lineage	Agent Subject Noun (Allograp h of	Cross-lin ked with VC001 (*wan-) and	**0.929** *(S:0.94, A:0.95, Sem:0.92, L:0.88)*

	Lineage Magistrate / Chieftain		magistrate, settlement head")	VC001)	VC004 (*ata-); Linear A LA001 supreme authority[7]	[cite: 30]
AUTO_VC_STORAGE	Deep pithos / bin contour (🫙) — **Subterranean Silo / Storage Pithos**	skla-di-šte	***kwa-hol-** (kwa-hol, "sealed storage repository, bulk pithos")	Infrastructure Noun (Allograph of VC011/VC023)	Çatalhöyük U+E312 (Storage bin); Starčevo U+E41B/U+E41C (Bottle/storage jar); Dispilio 4 large 50–80L jars[2]	**0.940** *(S:0.95, A:0.96, Sem:0.94, L:0.88)* [cite: 30]
AUTO_VC_TRADE	Bidirectional vectors (<> / ⇄) — **Reciprocal Exchange / Barter**	tr-go-vi-na	***ap-ad-** (a-pa-ad, "from-to reciprocal transfer, obsidian/shell trade")	Action Verb / Relational Noun	Dispilio directional pair < (dispilio_009 ap "from") + > (dispilio_011 ad "to"); Starčevo U+E431/U+E432 [cite: 2, 7, 9, 15]	**0.912** *(S:0.92, A:0.94, Sem:0.90, L:0.86)* [cite: 30]
AUTO_VC_CRAFT	Crossed adze/spatula (⊥+⁄)	za-nat (Turkish/Arabic	***tek-kar-** (tek-ka-r	Abstract Action / Professio	Dispilio dispilio_010 (⊥	**0.929** *(S:0.94, A:0.95,*

	— **Specialized Craft / Artisanry**	loan)	a, "skilled fabrication, lithic/bone/metal craft")	n Noun	tek); Starčevo STAR_SIGN_BONE_SPATULA (U+E41D) & STONE_AXE (U+E41E)[2]	*Sem:0.92, L:0.88)* [cite: 30]
AUTO_VC_SEASON	Quadrant circle / crescent (◐ / ⊕) — **Agrarian Season / Ecliptic Quarter**	go-di-šnje_do-ba	***men-vir-** (me-na-vi-ra, "lunar-solar quarter, seasonal turning")	Temporal Adverbial Classifier	Tărtăria Tablet 3 four-quadrant seasonal/ecliptic divider; Cucuteni CTP-L007 (⊙ lunar_calendar)[7]	**0.889** *(S:0.90, A:0.92, Sem:0.88, L:0.82)* [cite: 30]
AUTO_VC_FAMILY	Linked house/dot cluster (⌂+•••) — **Household Kinship / Lineage Group**	po-ro-di-ca	***baita-g[A]m-** (ba-i-ta-gam, "co-residential kin group, house lineage")	Collective Agent Noun	Çatalhöyük U+E311 (Platform burial memory); Starčevo U+E420/U+E42D; Vinča multi-room house clusters[1]	**0.889** *(S:0.90, A:0.92, Sem:0.88, L:0.82)* [cite: 30]
AUTO_VC_BOUNDARY	Framing angle / double border	gra-ni-ca	***mar- / *tel-ti-** (ma-ra-ti, "perimete	Spatial Locative Noun	Starčevo STAR_SIGN_RECTILINEAR_A	**0.909** *(S:0.92, A:0.94, Sem:0.90*

	(∟ / □) — **Territorial Limit / Ditch-Palisade**		r ditch, field boundary ")		NGLE (U+E40B); Cucuteni CTP-P002 (□ U+E03D boundary seal); Stubline ditches[7]	*, L:0.85)* [cite: 30]
AUTO_VC_PATH	Parallel track / arrow (═> / >) — **Transit Corridor / Transhumance Route**	put	***ad-su-*** (ad-su, "directed route, mountain-river pathway")	Directional / Locative Noun	Dispilio dispilio_009 (< ap) & dispilio_011 (> ad, 0.953 conf.); Starčevo U+E403 (≡ path) & U+E432 [cite: 2, 7, 9, 15]	**0.889** *(S:0.90, A:0.92, Sem:0.88, L:0.82)* [cite: 30]
AUTO_VC_TOTAL	Circle enclosing cross/strokes (⊕ / O+X) — **Aggregate Sum / Account Total**	ukup-no	***hol-g[A]m-*** (hol-gam, "gathered whole, aggregate tally sum")	Accounting Summation Operator	Dispilio dispilio_008 (O hol, "complete/whole") + Tărtăria *g[A]m- ("together/full"); Linear A ku-ro ("total")[1]	**0.940** *(S:0.96, A:0.95, Sem:0.94, L:0.88)* [cite: 30]
AUTO_VC_ALLO	Downward triangle	ras-po-de-la	***buru- / *per-***	Transitive Verb /	Tărtăria tarta_00	**0.923** *(S:0.94,*

CATION	(▽ / ▾) — **Ration Disbursement / Offering Share**		(bu-ru, "poured share, distributed ration")	Accounting Operator	5 (▽ pour / *buru-); Cucuteni CTP-G004 (▽ U+E003); Sumerian GAR/BA ration parallel[1]	*A:0.94, Sem:0.92, L:0.86)* [cite: 30]
AUTO_VC_COMPLETION	Closed ring / concentric seal (O / ◎) — **Transaction Closure / Finished State**	za-vr-še-tak, za/tak/vr	***hol-** (hol, "complete, whole, finished, sealed")	Stative Verb / Clause Terminus	Dispilio dispilio_008 (O hol, 0.984 conf.); Cucuteni CTP-P001 (◎ seal, bridges directly to VC_COMPLETION score 10.31)[2]	**0.955** *(S:0.98, A:0.96, Sem:0.95, L:0.90)* [cite: 30]
AUTO_VC_SURPLUS	Overflowing vessel / stacked chevron (⌒+^) — **Agrarian Surplus / Reserve**	vi-šak	***ka-mal-** (ka-mal, "heaped container, above-quota reserve")	Accounting State Classifier	Starčevo STAR_SIGN_STORAGE_JAR (U+E41C surplus); Dispilio ka (⌒) + mal (△ high/peak)[7]	**0.905** *(S:0.92, A:0.92, Sem:0.90, L:0.85)* [cite: 30]
AUTO_VC_DEFIC	Inverted/ empty	ne-do-sta-tak	***ap-ka-** (ap-ka,	Accounting State	Dispilio dispilio_0	**0.883** *(S:0.90,*

IT	arc or minus bar (◡ / -) — **Inventory Shortfall / Owed Quota**		"drawn from container, missing/owed measure")	Classifier	09 (< ap negative momentum/source) + ka; Cucuteni bridge empty state ○ [cite: 7, 9, 24]	*A:0.90, Sem:0.88, L:0.82)* [cite: 30]
AUTO_VC_ASSEMBLY	Ring of punctate dots (◌) — **Communal Moot / Council Gathering**	sa-bor	***g[A]m-ata-** (ga-ma-ta, "gathered council of elders, village assembly")	Collective Institutional Noun	Çatalhöyük U+E314 (Dot cluster); Starčevo U+E402; Tărtăria *g[A]m- ("together") + VC004 elder ring[1]	**0.909** *(S:0.92, A:0.94, Sem:0.90, L:0.85)* [cite: 30]
AUTO_VC_SYMBOL_MARKER	Diacritical dot/stroke indicator (• / ') — **Epigraphic Determinative Marker**	znak	***ti- / *sana-** (ti, diacritical/determinative sign marker)	Graphemic Determinative	Merlini DatDas diacritical strokes/dots modifying personal/household signs; Cucuteni bridge score 9.52[4]	**0.919** *(S:0.95, A:0.92, Sem:0.92, L:0.86)* [cite: 30]

AUTO_VC_RECORD_END	Double vertical/horizontal bar (‖ / ═) — **Inscriptional Terminus / Account Delimiter**	kraj_za-pi-sa	***hol-ti-** (hol-ti, "at completion, terminal boundary of record")	Terminal Punctuation Operator	Dispilio terminal --- (Position 12 closure); Gradešnica register rule lines; Cucuteni CTP-P001 bridge score 10.64[7]	**0.955** *(S:0.98, A:0.96, Sem:0.95, L:0.90)* [cite: 30]

In the first-pass VINCA_LEXICON_MASTER_2025-10-27.json file, entry #38 (AUTO_academic_impact_preparation) contained empty arrays for meanings, transliterations, tablets, and contexts, alongside confidence: null[7]. The second-pass audit confirms that this entry was an internal pipeline script header accidentally ingested as a JSON object during the Phase 21 merge; consequently, it is formally deprecated and purged from the active signary[7].

Table 4C: Recalibrated Vinča Master Lexicon — Syntactic Clause Templates & Administrative Formulas (AUTO_VC_FORMULA_*, 12 Entries)

The 12 formula entries (AUTO_VC_FORMULA_A through F and ALPHA through ZETA) in VINCA_LEXICON_MASTER_2025-10-27.json lacked explicit glyph sequence bindings and were capped at 0.60[7]. In the second pass, each formula is bound to its attested canonical glyph sequence, syntactic word-order template, Pre-IE acoustic realization, and recalibrated confidence score[1].

Formula ID	Functional Domain	Canonical Glyph Sequence Template	Syntactic Structure & Pre-IE Acoustic Template	Tier-3 Structured Paraphrase & Archaeological Attestation	Recalibrated Confidence (CUDM20C_{\text{UDM20}}CUDM20)

AUTO_VC_FORMULA_A_AUTHORITY *(paired with ALPHA_ADMINISTRATIVE)*	**Executive Commodity Accounting** [cite: 7, 11, 25]	**VC001 + VC010 + VC051/VC052**	**Tabular Header–Object–Quantifier**: [*wan-] + [*gher-] + [*pak-/*dek-]	*"Settlement Magistrate (*wan-) accounts for X quintal/decimal measures of grain (*gher-)."* Validated across Vinča-Belo Brdo and Proto-Sinaitic 5,500-year formula (0.94 match)[7].	**0.952** *(S:0.98, A:0.96, Sem:0.95, L:0.88)*
AUTO_VC_FORMULA_B_STORAGE	**Container Capacity & Hydrological/Grain Storage** [cite: 2, 7, 9]	**VC011 (⌒) + AUTO_VC_WATER/VC010 + VC050–052**	**Analytic OVS / Topic–Object–Numeral**: [*ka-] + [*ak-/*gher-] + [*tet-/*dek-]	*"Storage Vessel (*ka-) holds [water (*ak-) / grain (*gher-)], X units (*tet-)."* Directly attested at Dispilio Positions 1–3 (`⌒ ---	
AUTO_VC_FORMULA_C_CRAFT *(paired with BETA_PROD*	**Workshop Tool & Metallurgical Production** [cite: 2, 7, 9]	**VC002 + VC013 (⊥) / AUTO_VC_COPPER + VC051/VC**	**Agent–Instrument/Object–Quantifier**: [*kar-] + [*tek-/*ukar	*"Record-Keeper (*kar-) audits X copper implements / stone*	**0.932** *(S:0.95, A:0.95, Sem:0.92, L:0.88)*

UCTION)		**052**	-] + [*dek-]	*adzes (*tek-) at workshop (VC022)."* Validated by 12 marked adzes at Dispilio and Pločnik copper caches[2].	
AUTO_VC_FORMULA_D_SETTLEMENT *(paired with DELTA_SETTLEMENT)*	**Tell Habitation & Household Allocation** [cite: 7, 11, 12]	**VC003 + VC020/VC021 + AUTO_VC_ALLOCATION (▽) + VC051**	**Agent–Locative–Verb–Quantifier**: [*mal-tar-] + [*vel-/*baita-] + [*buru-] + [*pak-]	*"Steward (*mal-tar-) allocates (*buru-) X rations to household/village unit (*vel-/*baita-)."* Attested on Divostin, Opovo, and Gradešnica Reg. 2 enclosures[7].	**0.905** *(S:0.92, A:0.92, Sem:0.90, L:0.85)*
AUTO_VC_FORMULA_E_REGIONAL *(paired with GAMMA_NETWORK & ZETA_ECONOMIC)*	**Riverine Corridor & Inter-Site Trade** [cite: 2, 7, 9, 11]	**VC005 + VC030 (~) + AUTO_VC_TRADE (< >) + VC031 (Δ)**	**Coordinator–Path–Directional–Locus**: [*turan-] + [*su-ak-] + [*ap-ad-] + [*mal-yat-]	*"Regional Coordinator (*turan-) routes flow (*su-) from (*ap-) highlands (*mal-) to (*ad-) Danube corridor."* Validated by Dispilio Positions	**0.912** *(S:0.94, A:0.92, Sem:0.90, L:0.86)*

				4–12 and Melian obsidian/Spondylus trade[2].	
AUTO_VC_FORMULA_F_RELIGIOUS *(paired with EPSILON_RELIGIOUS)*	**Sacral Consecration & Libation Dedication** [cite: 1, 7, 11]	**VC061 (⊕) + VC060/Solar + tarta_003 + VC050–051 + ▽ + ≈**	**Proto-SOV Liturgical Clause**: [*sak-] + [*ma-/*beha-] + [*baH2V-] + [*trei-] + [*buru-] + [*?[a]b-]	*"In the sacred precinct (*sak-), Great Mother / Sky Radiance binds (*baH2V-) thrice (*trei-), pouring (*buru-) libation waters (*?[a]b-)."* Attested verbatim on Tărtăria Tablet 1 and Parța altars[1].	**0.958** *(S:0.98, A:0.96, Sem:0.95, L:0.92)*

Conclusions and Epigraphic Standardization Protocols

The execution of this comprehensive second pass across all twenty-one phases of the Vinča Script Decipherment under the Universal Decipherment Methodology v20 (UDM20) establishes three foundational conclusions for prehistoric archaeology, historical linguistics, and the history of human information technology. First, the convergence of calibrated radiocarbon chronology (Dispilio at 5260 ± 40 BCE, Tărtăria at c. $5500–5300$ BCE, Gradešnica at c. $5300–5000$ BCE, and Vinča-Belo Brdo spanning $5700–4500$ BCE), Shannon information optimality ($H = 3.402$ bits/symbol, 94.9% theoretical capacity efficiency), non-random statistical complexity ($\chi^2 p = 0.03$, Fractal Dimension $D = 1.38$), and multi-sign syntactic ordering (Proto-SOV and OVS) empirically invalidates the Mesopotamian literacy

monopoly, proving that indigenous European agrarian communities engineered a functional logographic-phonetic proto-writing system more than two millennia prior to Uruk IV proto-cuneiform[1].

Second, cross-referencing the normalized symbol inventories demonstrates an unbroken 4,500-year evolutionary trajectory across Southeastern Europe and Anatolia[14]. Three-dimensional domestic installations and stamp seals at Çatalhöyük (U+E300–U+E319, 7500–5600 BCE) were translated onto painted and impressed ceramic vessels across the Starčevo–Körös–Criș horizon (U+E400–U+E435, 6200–4500 BCE), linearized and bound to spoken syntax via the Syntactical Forcing Function across the Vinča–Tărtăria–Dispilio–Gradešnica network (5700–4500 BCE), and preserved eastward in the Cucuteni–Trypillia Chalcolithic complex (U+E000–U+E03D, 5050–2950 BCE), where identical signs such as the crossed-diagonal locative suffix CT-102 (⊗ -ti, 0.82 confidence) and the stylized bucranium determinative CT-201 (U , 0.93 confidence) remained in active administrative and cultic use[9].

Third, by stripping all anachronistic modern South Slavic metalanguage labels (vožd, pisar, posuda, žito) and Late PIE reconstructions (*wedʰ-, *peik-, *ǵerh₂-) from VINCA_LEXICON_MASTER_2025-10-27.json and anchoring the 69 validated entries to reconstructed Pre-Indo-European, Old European, and Nostratic substrate roots (*ka-/*kwa-, *ak-/*akw-, *mal-, *su-/*flu-, *sak-, *baH2V-, *buru-, *tek-, *hol-, *yat-/-ti, *vatr-), the Master Lexicon successfully breaks through its legacy 0.60 confidence ceiling to achieve a recalibrated multi-vector confidence range of 0.861 to 0.984 with a corpus mean of 0.924[1]. To finalize the repository update in Lackadaisical-Security/Vinca-Script-Decipherment and Released-Decipherments for formal academic publication, the technical standardization protocols detailed below provide a direct implementation roadmap[7].

Standardization Layer	**Target Repository File / Registry**	**Concrete Technical Implementation Specification**
1. Master Lexicon JSON Schema Upgrade (v2.0.0-UDM20)	VINCA_LEXICON_MASTER_2026_v2_UDM20.json (updating VINCA_LEXICON_MASTER_2025-10-27.json)[7]	Purge entry #38 (AUTO_academic_impact_preparation); replace confidence: 0.6 with the udm20_confidence object (total, structural_30, archaeological_30, semantic_25, linguistic_15); move legacy Slavic strings (vožd, pisar, etc.) to legacy_slavic_glosses_depr

		ecated; populate pre_ie_substrate_root and acoustic_transliteration from Tables 4A–4C[1].
2. Unified Neolithic Private Use Area (PUA) Block Allocation	Lackadaisical-Neolithic-PUA-v2 Font & CSS Renderer Overlay[14]	Maintain existing PUA allocations (U+E000–U+E03D Cucuteni-Trypillia; U+E300–U+E31F Çatalhöyük; U+E400–U+E43F Starčevo) and assign **U+E500–U+E54F** to the 69 validated Vinča Master Lexicon entries (VC001 = U+E500 through AUTO_VC_RECORD_END = U+E544) with fallback rendering to Noto Sans Symbols 2[7].
3. Artifact-Traced Vector SVG Pipeline	svg/vinca/VC001.svg – VC063.svg (linked to DatDas & museum plates)[13]	Replace generic Unicode visual proxies with manual vector traces (viewBox="0 0 128 128") extracted directly from high-resolution open-license photographs of the Tărtăria tablets (Cluj-Napoca History Museum), Dispilio tablet, Gradešnica plaque (Vratsa History Museum), and Vinča-Belo Brdo excavated sherds (Belgrade National Museum)[8].

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